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# DPRec: A dual-path preference evolution framework with semantic alignment for knowledge graph recommendation

Weihua Xu *, Linxi Yin

*College of Artificial Intelligence, Southwest University, ChongQing, 400715, China*

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ABSTRACT

Although recent advances in knowledge graph-based recommendation have achieved remarkable progress, several key challenges remain. First, most existing methods treat user preferences as static, failing to capture their semantic evolution over time. Second, many graph neural network models rely on uniform relation aggregation, overlooking differences in semantic importance and increasing the risk of over-smoothing. Furthermore, existing contrastive learning approaches primarily emphasize structural consistency by generating samples through random perturbations, which neglect preference hierarchies and can lead to semantic inconsistency. To address these limitations, this paper proposes DPRec, a dual-path preference evolution framework with semantic alignment for knowledge graph recommendation. DPRec jointly captures multi-hop user interests through direct and indirect preference path diffusion, incorporates a preference evolution attention mechanism for adaptive semantic aggregation, and employs a multi-level contrastive learning framework to align preference-level and path-level representations. Together, these components address three key aspects of knowledge graph recommendation, namely user preference modeling, dynamic preference evolution, and semantic alignment, thereby overcoming the limitations of existing approaches. Comprehensive experiments on multiple public datasets demonstrate that DPRec consistently outperforms existing baselines in terms of AUC, F1, Hit@K, and NDCG@K. These results demonstrate that dual-path preference reasoning produces more accurate, semantically consistent, and robust knowledge-aware recommendations.

## 1. Introduction

With the proliferation of digital technologies, the Internet has undergone significant transformation. A wide range of online platforms, including e-commerce websites, music streaming services, and social media networks, generate enormous amounts of content every day. As a result, users often struggle to discover information that matches their interests. Recommender systems (RSs) have emerged as an effective solution to this challenge by analyzing user behaviors and preferences to deliver personalized recommendations (Adomavicius & Tuzhilin, 2005; Zhang et al., 2019). Consequently, they reduce information overload and improve the overall user experience.

Early RSs primarily relied on content-based filtering (CBF) and collaborative filtering (CF) (Adomavicius & Tuzhilin, 2005). CBF estimates item similarity based on item attributes and users' historical behaviors, assuming that users prefer items with similar characteristics. In contrast, CF leverages user-item interactions and neighborhood similarity to identify users with comparable interests and generate recommendations (Koren et al., 2009; Sarwar et al., 2001). Matrix factorization (MF) later improved scalability by projecting users and items into a shared latent space, enabling effective modeling of implicit preference signals. Factorization machines (FMs) further extended this concept by modeling pairwise feature interactions within a unified factorization framework, demonstrating strong robustness under sparse conditions (Rendle, 2010). Despite these advances, traditional recommendation methods depend heavily on explicit ratings or dense interaction data and struggle to capture the rich semantic relationships among users, items, and their associated entities. Consequently, their performance deteriorates substantially in sparse data and cold-start scenarios (Koren et al., 2009; Lika et al., 2014). Incorporating external semantic knowledge into RSs has therefore become an important direction for addressing these limitations.

Knowledge graphs (KGs) have emerged as an effective source of external semantic information. By organizing entities and relationships as structured triples, KGs provide rich contextual knowledge and high-order connectivity that alleviate data sparsity while improving recommendation robustness and interpretability (Guan et al., 2025). Existing KG-based recommendation methods can generally be divided into

* Corresponding author.
*E-mail addresses:* chxuwh@swu.edu.cn (W. Xu), ylx06196@gmail.com (L. Yin).

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two categories based on how KG information is utilized: propagation-based methods and path reasoning-based methods (Guo et al., 2020). Propagation-based approaches aggregate neighborhood information into low-dimensional embeddings to capture multi-hop semantic relationships (Liu et al., 2021; Wang et al., 2021a). Path reasoning methods instead traverse relational paths and employ reinforcement learning or probabilistic inference to estimate semantic relevance, providing greater interpretability at the expense of increased computational complexity (Cui et al., 2022; Geng et al., 2022).

Although KG-based methods significantly enrich semantic information, their ability to model complex relational structures and high-order dependencies remains limited (Wang et al., 2022). Recent advances in graph neural networks (GNNs) have provided an effective framework for modeling user-item interactions on graph-structured data. Through message passing and neighborhood aggregation, GNNs capture high-order collaborative signals and contextual dependencies, producing more expressive and interpretable representations (He et al., 2020; Wu et al., 2021). To improve scalability and generalization on large graphs, recent studies have introduced inductive learning frameworks and neighborhood sampling strategies that enable efficient information propagation (Hamilton et al., 2017; Ying et al., 2018). Temporal extension modules further capture sequential and time-dependent user preferences (Chang et al., 2021; Huang et al., 2023), while attention mechanisms and multi-view architectures improve representation learning by adaptively weighting neighboring nodes and integrating multiple sources of contextual information (Faroughi et al., 2025; Wang et al., 2023). Collectively, these developments have extended GNN-based RSs beyond prediction accuracy toward improved reasoning capability, interpretability, and scalability.

Building on these advances, researchers have integrated GNNs with KGs to develop knowledge-enhanced recommendation frameworks. By incorporating external semantic structures and meta-path relationships, these hybrid models produce richer representations and stronger reasoning capabilities. Nevertheless, they continue to face challenges such as preference dilution, over-smoothing, and limited semantic adaptivity in complex heterogeneous graphs (Liu & Yang, 2023). More recently, self-supervised learning (SSL) and contrastive learning (CL) have been incorporated into graph-based recommendation to alleviate data sparsity, noisy interactions, and exposure bias. Multi-view contrastive frameworks align graph, hypergraph, and heterogeneous representations to learn more robust embeddings (Chen et al., 2023; Yu et al., 2023). Furthermore, knowledge-aware, social, and multi-channel extensions improve personalization (Liu & Yang, 2023; Yang et al., 2022), while disentanglement and debiasing techniques enhance robustness under sparse feedback (Mao et al., 2021; Ren et al., 2023). Together, these advances represent a shift from fully supervised recommendation pipelines toward self-supervised graph representation learning with stronger generalization capabilities (Yu et al., 2023, 2022).

Despite these advances, several important challenges remain. First, existing KG-based recommendation models primarily rely on knowledge propagation to enrich user and item representations, while the dynamic evolution of user preferences during semantic propagation remains insufficiently explored (Wang et al., 2018, 2019c, 2021a). In practice, user interests change over time and across contexts, requiring more adaptive preference modeling (Huang et al., 2023). Second, although GNN-based recommendation methods enable high-order information aggregation, they often fail to distinguish the varying semantic importance of heterogeneous neighboring information. As a result, irrelevant signals may accumulate during propagation, weakening personalized preference representations (He et al., 2020; Wang et al., 2019c,d). Third, existing CL methods primarily emphasize structural consistency (Wu et al., 2021; Yu et al., 2022) while overlooking preference-level semantic alignment. Most approaches generate positive and negative samples through random perturbations, ignoring fine-grained preference semantics and potentially producing structurally consistent but semantically mismatched representations (Wang et al., 2021b; Yang et al., 2022).

To address these challenges, we propose DPRec, a novel KG recommendation framework that jointly models multiple sources of user preferences and their dynamic evolution during multi-hop semantic propagation. Specifically, DPRec introduces a dual-path reasoning mechanism that disentangles user preferences into two complementary components: a direct preference set derived from user-item interactions and an indirect preference set inferred from collaborative user relationships. These complementary preference sets serve as the foundation for multi-hop semantic propagation over the KG, enabling the model to reason across both individual and collaborative semantic spaces. To capture evolving user interests, DPRec further incorporates a preference evolution attention mechanism that dynamically updates user preference representations at each propagation layer while adaptively weighting neighboring entities according to their semantic relevance. This layer-wise attention mechanism emphasizes highly relevant semantic information and suppresses noisy signals, producing more accurate representations of evolving user preferences. Finally, DPRec integrates preference-level and path-level CL through a dual-view InfoNCE objective to align structural and semantic representations, thereby improving semantic consistency and enhancing robustness under sparse and noisy conditions.

The primary contributions of this study are summarized as follows:

- **Dual-path preference-centric knowledge diffusion:** We propose a dual-path diffusion framework that initializes semantic propagation from complementary direct and indirect preference sets. By reasoning over both individual and collaborative semantic spaces, the framework learns richer and more expressive user preference representations.
- **Preference evolution and attention aggregation:** We develop a preference evolution attention mechanism that dynamically updates user preference representations and adaptively weights neighboring entities during graph propagation, reducing noise while improving semantic relevance.
- **Multi-level contrastive learning:** We introduce a unified CL framework that jointly models preference-level and path-level consistency. By aligning user preferences with semantic paths through a dual-view InfoNCE objective, the framework strengthens semantic alignment and improves representation robustness under sparse and noisy conditions.
- **Comprehensive evaluation:** Extensive experiments on multiple public datasets demonstrate that DPRec consistently outperforms state-of-the-art KG- and GNN-based recommendation models in terms of AUC, F1, and Hit@K, validating both its effectiveness and interpretability.

## 2. Related work

### 2.1. Traditional recommendation methods

Traditional recommendation methods primarily rely on CBF and CF to capture user preferences. Sarwar et al. (2001) proposed a neighborhood-based algorithm that predicts user preferences based on user or item similarity. Building on this work, Koren et al. (2009) introduced a MF framework that projects users and items into a shared latent space and further enhanced it through SVD++ by incorporating implicit feedback. Rendle (2010) subsequently generalized this idea with FMs, which model pairwise feature interactions with linear computational complexity, improving scalability for industrial applications. Despite their success, these methods rely heavily on explicit ratings and are unable to capture high-order or contextual relationships among users and items. Consequently, research has gradually shifted from latent factor models toward graph-based learning approaches, which better model structural dependencies and semantic relationships in complex recommendation scenarios.

### 2.2. Knowledge graph-based recommendation methods

KGs have been widely incorporated into RSs to alleviate the data sparsity and cold-start problems inherent in CF. By organizing entities and relations into structured triples, KGs provide rich external semantic information that enhances user-item representations. Existing KG-based recommendation methods can generally be categorized into three groups: embedding-based methods, path reasoning methods, and hybrid methods.

Embedding-based methods learn low-dimensional representations of entities and relations. Hui et al. (2022) proposed ReBKC, which combines KG embeddings with behavioral features through a self-attention mechanism. Sha et al. (2021) developed HAKG to encode hierarchical subgraphs and capture both semantic and topological information. Li et al. (2021) introduced KG4Vis, which combines TransE with self-adversarial sampling to reduce the influence of noisy triplets. Although effective, these methods generally have limited ability to model explicit multi-hop semantic relationships.

Path reasoning methods, including RSL-GRU (Cui et al., 2022), ReMR (Wang et al., 2022), and TPRec (Zhao et al., 2022), explicitly explore multi-hop relational paths through reinforcement learning or temporal reasoning, thereby improving recommendation interpretability. However, their reliance on path exploration often results in relatively high computational costs.

To balance the strengths of these approaches, hybrid methods combine embedding learning with path-aware reasoning. Representative models such as KGIN (Wang et al., 2021a) and CG-KGR (Chen et al., 2022) integrate embedding representations with path-aware aggregation to improve both recommendation accuracy and interpretability. Wang et al. (2024) further enhanced knowledge-aware representation learning by introducing fine-grained attention and contrastive KG embedding, reducing the influence of redundant semantic information. More recently, diffusion-based approaches have extended KG recommendations by incorporating multi-view diffusion mechanisms to capture complementary semantic information. For example, Li et al. (2026b) proposed a dual-view diffusion framework that introduces noisy and structural diffusion processes from different perturbation perspectives. By leveraging complementary diffusion views, the framework alleviates the over-denoising problem while preserving task-relevant semantic information during KG refinement.

### 2.3. Graph neural network-based recommendation methods

While KG-based methods provide rich semantic information, their ability to model complex relational structures and perform efficient multi-hop aggregation remains limited. To address these limitations, GNNs have been introduced to enhance structural modeling and representation learning. In particular, GNNs provide a unified mechanism for propagating and integrating semantic information from KGs through message passing, making them a natural complement to KG-based methods. KGs provide structured semantic information, whereas GNNs leverage this structure for multi-hop propagation and representation learning. As GNNs have advanced, researchers have increasingly incorporated graph structures into RSs to better capture high-order associations between users and items.

Wang et al. (2019d) proposed NGCF, which models high-order connections on user-item graphs, while Wang et al. (2020a) introduced DGCF to disentangle latent intents and improve interpretability. To alleviate over-smoothing, He et al. (2020) designed LightGCN by removing nonlinear activation functions and feature transformations. For improved scalability, GraphSAGE (Hamilton et al., 2017) and PinSage (Ying et al., 2018) adopt neighborhood sampling strategies for large-scale industrial recommendation. In sequential recommendation, Chang et al. (2021) and Huang et al. (2023) incorporated temporal signals to model dynamic user preferences.

To enhance semantic reasoning, Fu et al. (2020) proposed MAGNN with meta-path aggregation for heterogeneous graphs, while Hu et al. (2023) developed CGCL to incorporate social influence. In addition, Wang et al. (2019b) and Cao et al. (2019) leveraged KGs to improve semantic propagation and transfer learning. To capture fine-grained user interests, Zhang et al. (2023) proposed MA-GNNs, which construct aspect-aware graphs from review sentiments and fuse them through a routing mechanism. Similarly, Li et al. (2025) proposed a user preference interaction fusion strategy and a user preference swap attention mechanism, which incorporate user preference information and assign importance weights to neighboring entities, thereby enhancing GNN-based personalized recommendation.

More recently, researchers have explored diffusion-based approaches for social and graph-based recommendation. For example, He et al. (2026) introduced diffusion processes into social recommendation by performing denoising and reconstruction in the latent space, thereby reducing the influence of noisy or irrelevant social connections and improving recommendation performance. Overall, these methods have significantly advanced recommender systems by improving representation learning, semantic reasoning, and robustness, highlighting the potential of graph learning paradigms for intelligent recommendation.

### 2.4. Graph contrastive learning methods

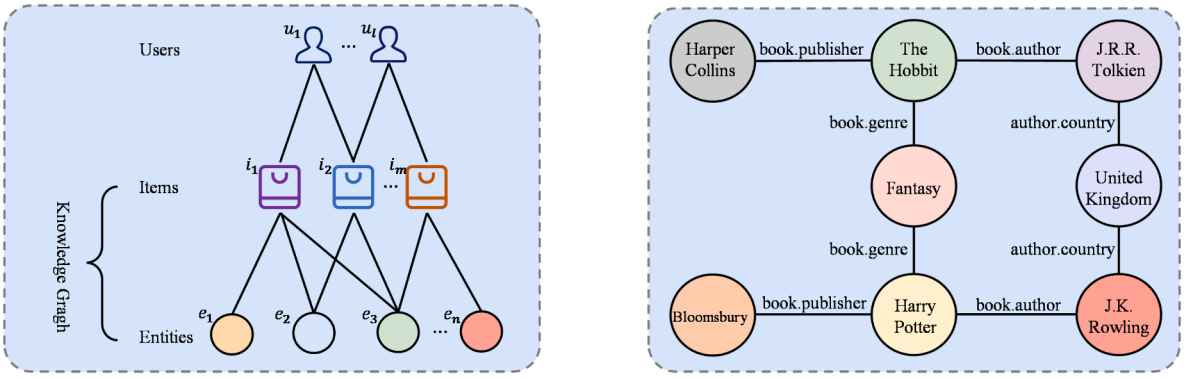
Despite the strong representation learning capability of GNN-based recommendation models, they remain vulnerable to data sparsity, noise, and limited generalization. To address these challenges, researchers have introduced CL as an effective self-supervised paradigm for improving representation robustness and semantic consistency. Wu et al. (2021) proposed SGL, which applies the InfoNCE loss to randomly perturbed user-item graphs to improve robustness. Building on this work, Cai et al. (2023) developed LightGCL, which constructs low-rank global views without explicit augmentation, while Xia et al. designed HCCF using hypergraph constraints to capture higher-order signals. Yu et al. (2023, 2022) simplified augmentation through noise injection and unified encoding in XSimGCL. Ren et al. (2023) proposed DCCF with disentangled multi-intent learning, while Mao et al. improved embedding uniformity using cosine contrastive loss. Beyond bipartite graphs, Liu and Yang (2023) introduced MGCL for social influence modeling, whereas Wang et al. (2021b) and Yang et al. (2022) extended CL to heterogeneous graphs and KGs, respectively. Li et al. (2026a) further incorporated a hierarchy-aware diffusion model into knowledge-enhanced CL to address knowledge imbalance and improve recommendation performance.

Although existing KG-based, GNN-based, and CL-based recommendation methods have achieved significant progress, several limitations remain. KG-based methods lack explicit modeling of preference evolution. GNN-based methods often fail to distinguish the varying semantic importance of heterogeneous information during propagation, weakening personalized preference modeling. CL-based methods primarily emphasize consistency across augmented views, while fine-grained preference-level semantic alignment remains insufficiently explored. These limitations motivate the proposed DPRec framework, which jointly models complementary user preferences, dynamic preference evolution, and semantic alignment.

## 3. Problem formulation

We formulate the personalized recommendation problem that leverages external knowledge as follows. Let $\mathcal{U} = \{u_1, u_2, \dots, u_M\}$ be the set of users and $\mathcal{I} = \{i_1, i_2, \dots, i_N\}$ be the set of items. The historical interactions between users and items can be represented as a binary matrix $\mathbf{Y} = [y_{ui}] \in \mathbb{R}^{M \times N}$, where

$$y_{ui} = \begin{cases} 1, & \text{if user } u \text{ has interacted with item } i; \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$




(a) Unified user–item–entity structure

(b) Example of the book knowledge graph

**Fig. 1.** Illustration of the knowledge graph structure used in our framework.

Each positive interaction ($y_{ui} = 1$) indicates that user $u$ shows a behavioral preference for item $i$. Due to the sparsity of feedback data, it is difficult to fully infer user interests based on $\mathbf{Y}$ alone.

**Knowledge graph.** To enrich the semantic information of items, we introduce an external knowledge graph $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$, where $\mathcal{E}$, $\mathbf{R}$, and $\mathcal{T}$ denote the sets of entities, relations, and triples, respectively. Each triple $(h, r, t) \in \mathcal{T}$ expresses a factual relation $r$ from a head entity $h$ to a tail entity $t$. For each item $i \in \mathcal{I}$, a mapping function $\phi : \mathcal{I} \to \mathcal{E}$ is defined to associate the item with one or more entities in the knowledge graph. This mapping aligns the recommendation space with structured semantic knowledge, allowing the model to capture multi-relational dependencies beyond direct interactions.

As shown in Fig. 1a, the proposed framework adopts a unified structure that integrates user–item interactions with item–entity relations, enabling joint modeling of collaborative and semantic information. Fig. 1b provides a concrete example from the book domain, demonstrating how semantic relations link different entities. This structured knowledge provides valuable context for preference modeling and enables multi-hop reasoning across semantic paths.

**Recommendation task.** Given the interaction matrix $\mathbf{Y}$ and the knowledge graph $\mathcal{G}$, our goal is to estimate user preferences for unobserved items. The predictive function is defined as follows:

$$\hat{y}_{ui} = \mathcal{F}(u, i \mid \mathbf{Y}, \mathcal{G}, \Pi), \tag{2}$$

where $\hat{y}_{ui}$ indicates the interaction likelihood, and $\Pi$ denotes the set of learnable parameters. The function $\mathcal{F}(\cdot)$ jointly models collaborative signals from user–item interactions and semantic dependencies from the knowledge graph. In our framework, $\mathcal{F}$ is implemented as a dual-path, preference evolution graph neural network that integrates explicit and implicit user preferences through multi-hop propagation over $\mathcal{G}$.

## 4. Methodology

### 4.1. Overall framework

Fig. 2 illustrates the overall architecture of the proposed DPRec model, which consists of the following key components: (A) Dual-path preference diffusion module; (B) User preference interaction encoding module; (C) Preference evolution attention aggregation; (D) Preference-level contrast; (E) Path embedding module; (F) Path-level contrast; (G) Final prediction. The proposed framework integrates user-item interactions with KG semantics to jointly model explicit and implicit user preferences within a unified reasoning framework.

### 4.2. Dual-path preference-centric knowledge diffusion

#### 4.2.1. User-side dual-path preference diffusion

On the user side, our model establishes a dual-path diffusion mechanism to capture both direct and indirect components of user preferences. While existing KG-based recommendation methods typically treat user preferences as a unified signal during graph propagation, our approach explicitly separates preferences derived from direct user–item interactions and those inferred from collaborative user behaviors. This design enables the model to represent both explicitly expressed preferences and potential preferences inferred from collaborative signals, resulting in a more fine-grained semantic representation of user interests. The dual-path mechanism provides the initialization for subsequent preference diffusion over the KG.

The first path, termed the *Direct Path*, models explicitly expressed user preferences. For each user $u$, we collect the set of items with which the user has positively interacted (e.g., clicks, likes, or ratings). These items are then mapped to their corresponding entities in the KG, which serve as the seed nodes for semantic diffusion, defined as:

$$\mathcal{E}_u^{\text{dir}} = \{v_i \mid i \in \mathcal{I}_u^+, v_i = \phi(i)\}, \tag{3}$$

where $\mathcal{I}_u^+$ denotes the set of positive-feedback items for user $u$, and $\phi(i)$ is the mapping function linking item $i$ to its corresponding entity in the KG. The set $\mathcal{E}_u^{\text{dir}}$ represents the user's confirmed interests and provides the foundation for the subsequent semantic diffusion process.

The second path, termed the *Indirect Path*, expands user preferences through collaborative neighbors. This approach is based on the assumption that users with overlapping interaction histories are likely to share similar interests. For a target user $u$, we identify neighboring users $u'$ who share common interactions with $u$. We then aggregate the items with which these neighbors have positively interacted and map them to their corresponding KG entities. The resulting indirect preference set is formalized as:

$$\mathcal{E}_u^{\text{ind}} = \{v_j \mid \exists u' \neq u, \mathcal{I}_u^+ \cap \mathcal{I}_{u'}^+ \neq \emptyset, j \in \mathcal{I}_{u'}^+, v_j = \phi(j)\}, \tag{4}$$

where $\mathcal{E}u^{\text{ind}}$ denotes the set of indirect preferences for user $u$, inferred from the behaviors of similar users $u'$. Here, $\mathcal{I}_{u'}^+$ denotes the set of positive-feedback items associated with neighboring user $u'$. By incorporating collaborative behavioral information, this indirect preference set expands the user's semantic space and enables the model to uncover potential interests that are not evident from explicit interactions alone.

Both $\mathcal{E}_u^{\text{dir}}$ and $\mathcal{E}_u^{\text{ind}}$ serve as seed sets for diffusion in the KG. The direct path captures users' confirmed behavioral interests, whereas the indirect path supplements them with complementary semantic information derived from similar users. Together, the two paths enable the model to learn richer user preference representations and improve generalization under sparse interaction conditions.

#### 4.2.2. Item-side dual-path semantic diffusion

On the item side, our model adopts a symmetrical dual-path diffusion mechanism. This design enables each item to capture not only its inherent semantic meaning but also an enriched semantic context derived from user behaviors.

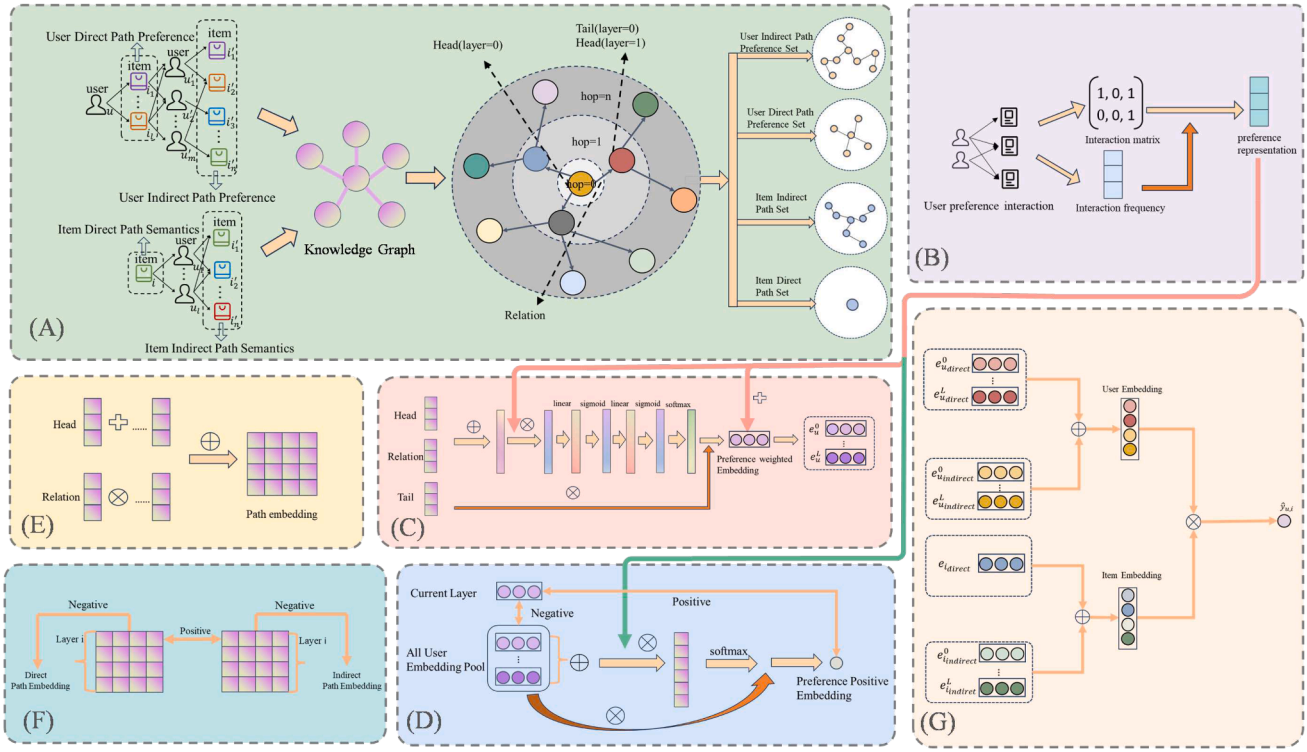



**Fig. 2.** Overview of the DPRec framework.

For each item $i$, if its entity $v_i$ is present in the KG, we initialize the direct semantic set as follows:

$$\mathcal{E}_i^{\text{dir}} = \{v_i \mid v_i = \phi(i)\}. \tag{5}$$

This set preserves the intrinsic knowledge-based semantics of the item, providing a stable semantic anchor for subsequent diffusion.

The *Indirect Semantic Set* enriches item semantics through user interactions. For item $i$, we retrieve the set of users $\mathcal{U}_i$ who have interacted with it. For each user $u \in \mathcal{U}_i$, we collect the entities corresponding to all items in the user's interaction history. These entities are then merged to form the indirect semantic set:

$$\mathcal{E}_i^{\text{ind}} = \bigcup_{u \in \mathcal{U}_i} \{v_j \mid j \in \mathcal{I}_u^+, v_j = \phi(j)\}. \tag{6}$$

This indirect semantic set provides each item with additional semantic neighbors derived from collaborative user behaviors, facilitating the learning of shared semantic patterns among items that frequently co-occur in user interactions.

Similarly, both $\mathcal{E}_i^{\text{dir}}$ and $\mathcal{E}_i^{\text{ind}}$ serve as seed sets for semantic diffusion. Together, they enable each item to capture both intrinsic and context-aware semantics, thereby enriching item representations and improving their generalization capability.

### 4.2.3. Multi-hop knowledge diffusion

Following the construction of the preference entity sets, the model performs a multi-hop diffusion process over the KG. This process propagates user preferences and item semantics along relational paths, capturing deeper structural dependencies and latent semantic relationships.

For each user $u$, the direct and indirect preference entities are maintained as two diffusion views: $\mathcal{E}_u^{\text{dir},(0)} = \mathcal{E}_u^{\text{dir}}$ and $\mathcal{E}_u^{\text{ind},(0)} = \mathcal{E}_u^{\text{ind}}$. Similarly, for each item $i$, two semantic diffusion views are initialized as $\mathcal{E}_i^{\text{dir},(0)} = \mathcal{E}_i^{\text{dir}}$ and $\mathcal{E}_i^{\text{ind},(0)} = \mathcal{E}_i^{\text{ind}}$. At each layer $l$, the model samples a fixed number $K$ of triples $(h, r, t)$ connected to entities from the preceding layer for each diffusion view. This sampling strategy mitigates path explosion while preventing overfitting to densely connected regions of the graph. The propagation rule is defined as:

$$\mathcal{T}_x^{v,(l)} = \{(h, r, t) \mid h \in \mathcal{E}_x^{v,(l-1)},\ (h, r, t) \in \mathcal{G}\}, \tag{7}$$

where $x \in \{u, i\}$ denotes the user or item side, and $v \in \{\text{dir}, \text{ind}\}$ represents the direct or indirect diffusion view. $\mathcal{T}_x^{v,(l)}$ denotes the set of triples sampled at layer $l$ for view $v$, while $\mathcal{E}_x^{v,(l-1)}$ represents the entity set obtained from the previous diffusion layer. The diffusion process proceeds independently for the two views until the predefined depth $L$ is reached. Consequently, direct behavioral preferences and indirect collaborative semantics propagate along separate relational paths while preserving their complementary characteristics.

The embeddings generated at each diffusion layer are retained for subsequent layer-wise aggregation. By aggregating representations across different propagation depths, the model combines local and global semantic information to generate high-order user and item representations. Finally, the representations learned from the two diffusion views are fused to obtain comprehensive user and item embeddings. As shown in Fig. 2(A), this module forms a core component of the proposed framework by integrating user-item interaction information with KG propagation through dual diffusion paths. The overall diffusion procedure is summarized in Algorithm 1. The time complexity of the algorithm is $O(2(|\mathcal{U}| + |\mathcal{I}|)LK)$, and the space complexity is $O(2(|\mathcal{U}| + |\mathcal{I}|)LK)$, where $|\mathcal{U}|$ and $|\mathcal{I}|$ denote the numbers of users and items, respectively, $L$ represents the number of propagation layers, and $K$ is the number of sampled triples at each propagation layer. The factor 2 corresponds to the direct and indirect preference views.

### 4.3. Refining representations through preference evolution and attention aggregation

Traditional recommendation methods typically treat user preferences as static features, making it difficult to capture the dynamic evolution of user interests. To address this limitation, as illustrated in Fig. 2(B)–(C), we introduce a Preference evolution and attention aggregation mechanism that leverages multi-hop propagation over KGs to

**Algorithm 1** Preference-centric knowledge diffusion.

**Input:** Interaction matrix $\mathbf{R}$, KG $\mathcal{G}$, mapping $\phi(\cdot)$, layers $L$, sample size $K$.
**Output:** Triple sets $\{\mathcal{T}_x^v\}_{x\in\{u,i\},v\in\{\text{dir,ind}\}}$.

// **1. Initialize Dual-Path Entity Sets**
**for** each user $u \in \mathcal{U}$ **do**
  Construct direct preference set:
  $\mathcal{E}_u^{\text{dir}} = \{v_i \mid i \in \mathcal{I}_u^+,\ v_i = \phi(i)\}$;
  Construct indirect preference set:
  $\mathcal{E}_u^{\text{ind}} = \{v_j \mid \exists u' \neq u,\ \mathcal{I}_u^+ \cap \mathcal{I}_{u'}^+ \neq \emptyset,\ j \in \mathcal{I}_{u'}^+,\ v_j = \phi(j)\}$;
  Initialize two diffusion views:
  $\mathcal{E}_u^{\text{dir},(0)} = \mathcal{E}_u^{\text{dir}}$;
  $\mathcal{E}_u^{\text{ind},(0)} = \mathcal{E}_u^{\text{ind}}$;
**end for**
**for** each item $i \in \mathcal{I}$ **do**
  Construct direct semantic set:
  $\mathcal{E}_i^{\text{dir}} = \{v_i \mid v_i = \phi(i)\}$;
  Construct indirect semantic set:
  $\mathcal{E}_i^{\text{ind}} = \bigcup_{u\in\mathcal{U}_i}\{v_j \mid j \in \mathcal{I}_u^+,\ v_j = \phi(j)\}$;
  Initialize two diffusion views:
  $\mathcal{E}_i^{\text{dir},(0)} = \mathcal{E}_i^{\text{dir}}$;
  $\mathcal{E}_i^{\text{ind},(0)} = \mathcal{E}_i^{\text{ind}}$;
**end for**
// **2.Multi-Hop Diffusion**
**for** each node type $x \in \{u, i\}$ **do**
  **for** each preference view $v \in \{\text{dir}, \text{ind}\}$ **do**
    Initialize: $\mathcal{T}_x^v = \emptyset$;
    **for** $l = 1$ to $L$ **do**
      Generate candidate triples from previous layer:
      $\mathcal{T}_c^{v,(l)} = \{(h,r,t) \mid h \in \mathcal{E}_x^{v,(l-1)}, (h,r,t) \in \mathcal{G}\}$;
      Sample $K$ triples:
      $\mathcal{T}_x^{v,(l)} \subseteq \mathcal{T}_c^{v,(l)}$;
      Update entities:
      $\mathcal{E}_x^{v,(l)} = \{t \mid (h,r,t) \in \mathcal{T}_x^{v,(l)}\}$;
      Accumulate triples:
      $\mathcal{T}_x^v = \mathcal{T}_x^v \cup \mathcal{T}_x^{v,(l)}$;
    **end for**
  **end for**
**end for**
**return** $\{\mathcal{T}_x^v\}_{x\in\{u,i\},v\in\{\text{dir,ind}\}}$;

iteratively refine user and item representations. This mechanism computes attention weights based on dynamically updated preference representations, overcoming the limitations of relying solely on structural features while enabling adaptive and personalized information aggregation at each propagation layer. Through iterative attention aggregation, the model progressively refines user preference representations, allowing them to adapt to evolving user interests.

First, we derive the initial preference representation from user interactions using a frequency-weighted nonlinear transformation:

$$\mathbf{p}_u^{(0)} = f(\mathbf{X}_u\mathbf{W}_s + \mathbf{b}_s) \cdot freq \tag{8}$$

where $\mathbf{p}_u^{(0)}$ denotes the initial explicit preference representation, $f(\cdot)$ is the nonlinear activation function, $\mathbf{X}_u$ is the user interaction feature matrix, $\mathbf{W}_s$ and $\mathbf{b}_s$ are the learned parameters, and $freq$ represents the interaction frequency weight between users and items.

Building upon this initial representation, we introduce a preference evolution attention mechanism at each propagation layer to account for dynamically changing item contributions. For a user $u$ and the $k$th interaction triple, the interaction representation is computed as:

$$\mathbf{z}_{u,k}^{(l)} = (\mathbf{h}_{u,k}^{(l)} \parallel \mathbf{r}_{u,k}^{(l)}) \cdot \mathbf{p}_u^{(l-1)}, \tag{9}$$

where $\mathbf{h}_{u,k}^{(l)}$ and $\mathbf{r}_{u,k}^{(l)}$ denote the embedding vectors of the head entity and relation of the $k$th triple at layer $l$, respectively. $\parallel$ denotes vector concatenation, and $\mathbf{z}_{u,k}^{(l)}$ is the weighted interaction representation for the corresponding triple. This formulation allows the current user preference representation to guide feature aggregation at each propagation layer.

The representation $\mathbf{z}_{u,k}^{(l)}$ is then passed through a neural network to compute the attention weight:

$$\text{att_score}_{u,k}^{(l)} = \text{Sigmoid}\Big(\text{Linear}\Big(\text{Sigmoid}\Big(\text{Linear}\Big(\mathbf{z}_{u,k}^{(l)}\Big)\Big)\Big)\Big), \tag{10}$$

where the two linear layers and sigmoid activation functions compute the attention score. The output $\text{att_score}_{u,k}^{(l)}$ represents the unnormalized attention weight for the $k$th triple of user $u$ at layer $l$.

The attention scores are then normalized using the softmax function to ensure that the weights sum to 1:

$$\alpha_{u,k}^{(l)} = \frac{\exp(\text{att_score}_{u,k}^{(l)})}{\sum_{j=1}^{T}\exp(\text{att_score}_{u,j}^{(l)})}, \tag{11}$$

where $T$ denotes the size of the user's sampled triple set.

The model then performs weighted aggregation over all tail entities:

$$\mathbf{e}_u^{(l)} = \sum_{k=1}^{T} \alpha_{u,k}^{(l)} \cdot \mathbf{t}_{u,k}^{(l)}, \tag{12}$$

where $\mathbf{e}_u^{(l)}$ represents the embedding of user $u$ at layer $l$ after aggregation, and $\mathbf{t}_{u,k}^{(l)}$ is the embedding of the tail entity corresponding to the $k$th triple. This weighted aggregation enables the model to integrate information from multiple relational neighbors while suppressing noisy or irrelevant entities.

To capture evolving user interests, the preference evolution mechanism iteratively updates the user preference vector by incorporating aggregated semantic information:

$$\mathbf{p}_u^{(l)} = \mathbf{p}_u^{(l-1)} + \mathbf{e}_u^{(l)}. \tag{13}$$

Similarly, the item representation $\mathbf{e}_i^{(l)}$ is computed symmetrically. At each propagation layer, the item's representation is updated through the preference evolution attention mechanism, allowing it to reflect both its intrinsic semantics and evolving user preferences. Consequently, the learned item representation incorporates both item-specific characteristics and contextual preference information, enabling more personalized recommendations.

To capture complementary semantic information across different propagation depths, we consider multiple aggregation strategies, including summation, max-pooling, and concatenation, allowing the model to flexibly integrate representations from different layers. Let $\mathbf{e}_x^{(l)}$ represent the embedding of user or item $x$ at layer $l$ ($x \in \{u, i\}$). The aggregated representation for both direct and indirect views is computed as:

$$\mathbf{e}_x^{\text{view}} = \text{AGG}\big(\mathbf{e}_x^{(1)}, \mathbf{e}_x^{(2)}, \ldots, \mathbf{e}_x^{(L)}\big), \tag{14}$$

where view $\in \{\text{dir}, \text{ind}\}$. We investigate three specific instantiations of the aggregation function $\text{AGG}(\cdot)$:

- **DPRec-Sum**: Utilizes element-wise summation, defined as $\text{AGG}(\cdot) = \sum_{l=1}^{L}\mathbf{e}_x^{(l)}$, to accumulate semantic signals from all layers, assuming that features reinforce each other across layers.
- **DPRec-Pool**: Employs element-wise max-pooling, defined as $\text{AGG}(\cdot) = \max_{l=1}^{L}(\mathbf{e}_x^{(l)})$, to select the most salient features while filtering out weaker signals.
- **DPRec-Concat**: Concatenates representations from all layers, defined as $\text{AGG}(\cdot) = \mathbf{e}_x^{(1)} \oplus \mathbf{e}_x^{(2)} \oplus \cdots \oplus \mathbf{e}_x^{(L)}$, to preserve complete hierarchical information without compression.

This aggregation process integrates shallow and deep semantic information from both the direct and indirect views, yielding more stable and comprehensive user and item representations. By combining local interaction information with higher-order relational semantics, the final representations improve both recommendation accuracy and robustness. The procedure is detailed in Algorithm 2. The time complexity of the algorithm is $O\big(2(|\mathcal{U}| + |\mathcal{I}|)LKd^2\big)$, and the space complexity

is $O(2(|\mathcal{U}|+|\mathcal{I}|)LKd)$, where $|\mathcal{U}|$ and $|\mathcal{I}|$ denote the numbers of users and items, respectively, $L$ represents the number of propagation layers, $K$ denotes the number of sampled triples at each propagation layer, and $d$ is the embedding dimension. The factor 2 corresponds to the direct and indirect preference views.

**Algorithm 2** Preference evolution and attention aggregation.

**Input:** User features $\{\mathbf{X}_u\}$, Triple sets $\mathcal{T}_u^{\text{dir/ind}}$ and $\mathcal{T}_i^{\text{dir/ind}}$, Layers $L$, Params $\Theta$.
**Output:** Final representations $\mathbf{e}_u^{\text{final}}, \mathbf{e}_i^{\text{final}}$.

// **1. Initialize Preference**
**for** each user $u$ **do**
  $\mathbf{p}_u^{(0)} = f(\mathbf{X}_u\mathbf{W}_s + \mathbf{b}_s) \cdot freq$;
**end for**
// **2. Multi-view Preference Evolution**
**for** each user $u$ **do**
  **for** each view $\in \{\text{dir}, \text{ind}\}$ **do**
    Initialize $\mathbf{p}_u = \mathbf{p}_u^{(0)}$;
    **for** $l = 1$ to $L$ **do**
      Let $\mathcal{T} = \mathcal{T}_u^{\text{view},(l)}$;
      *Compute interaction & attention scores for all* $k \in \mathcal{T}$*:*
      $\mathbf{z}_{u,k}^{(l)} = (\mathbf{h}_{u,k}^{(l)} \,\|\, \mathbf{r}_{u,k}^{(l)}) \cdot \mathbf{p}_u^{(l-1)}$;
      $\text{att_score}_{u,k}^{(l)} = \sigma\big(\text{Linear}\big(\sigma(\text{Linear}(\mathbf{z}_{u,k}^{(l)}))\big)\big)$;
      *Normalize & Aggregate:*
      $\alpha_{u,k}^{(l)} = \frac{\exp(\text{att_score}_{u,k}^{(l)})}{\sum_{j=1}^{T}\exp(\text{att_score}_{u,j}^{(l)})}$;
      $\mathbf{e}_{u,\text{view}}^{(l)} = \sum_{k=1}^{T} \alpha_{u,k}^{(l)} \cdot \mathbf{t}_{u,k}^{(l)}$;
      *Update preference:* $\mathbf{p}_u^{(l)} = \mathbf{p}_u^{(l-1)} + \mathbf{e}_{u,\text{view}}^{(l)}$;
    **end for**
  **end for**
**end for**
// **3. Final Fusion**
**for** each user $u$ **do**
  $\mathbf{e}_u^{\text{dir}} = \text{AGG}(\{\mathbf{e}_{u,\text{dir}}^{(l)}\}_{l=1}^{L}), \quad \mathbf{e}_u^{\text{ind}} = \text{AGG}(\{\mathbf{e}_{u,\text{ind}}^{(l)}\}_{l=1}^{L})$;
  $\mathbf{e}_u^{\text{final}} = \mathbf{e}_u^{\text{dir}} \oplus \mathbf{e}_u^{\text{ind}}$;
**end for**
// *Perform symmetrical operations for items to obtain* $\mathbf{e}_i^{\text{final}}$
**return** $\mathbf{e}_u^{\text{final}}, \mathbf{e}_i^{\text{final}}$.

### 4.4. Preference-level contrast

To enhance the consistency between structural and semantic representations, we design a preference-level CL module. Unlike conventional methods that construct contrastive pairs through random perturbations, this module leverages preference-aware representations to guide the formation of positive and negative samples, enabling more semantically meaningful alignment. As illustrated in Fig. 2(B)-(D), the module aligns user embeddings with preference-level positive samples while distinguishing them from irrelevant negatives, providing global semantic supervision that stabilizes representation learning.

For each training mini-batch, we collect user structural representations from all propagation layers and concatenate them to form the candidate pool:

$$\mathcal{E}^{pool} = \{\mathbf{e}_k^{pool}\}_{k=1}^{N}, \tag{15}$$

where $\mathbf{e}_k^{pool} \in \mathbb{R}^d$ denotes the $k$th candidate embedding, and $N$ is the total number of candidate embeddings constructed from the current mini-batch. The candidate pool contains user representations from all propagation layers and is dynamically updated for each mini-batch during training. For each anchor representation, its own embedding is excluded from the candidate pool when constructing the positive proxy and contrastive references.

Given the candidate pool, the similarity between the initial user preference vector and each candidate embedding is computed using their dot product:

$$s_{u,k}^{(l)} = \mathbf{p}_u^{(0)\top}\mathbf{e}_k^{pool}, \tag{16}$$

where $\mathbf{p}_u^{(0)} \in \mathbb{R}^d$ is the initial user preference vector derived from the user's explicit interaction history, $\mathbf{e}_k^{pool}$ denotes the $k$th candidate embedding in the pool, and $s_{u,k}^{(l)}$ represents the similarity between the initial preference vector and the $k$th candidate embedding at propagation layer $l$. This preference-guided similarity provides semantic guidance during propagation, encouraging the learned representations to remain aligned with the user's underlying interests while reducing semantic drift.

The similarity scores are then converted into a soft weighting distribution using a temperature-scaled softmax function:

$$\alpha_{u,k}^{(l)} = \frac{\exp(s_{u,k}^{(l)}/\tau)}{\sum_{\mathbf{e}_m^{pool}\in\mathcal{E}^{pool}\setminus\{\mathbf{e}_u^{(l)}\}}\exp(s_{u,m}^{(l)}/\tau)}, \tag{17}$$

where $\tau > 0$ is the temperature parameter controlling the sharpness of the distribution, and $\alpha_{u,k}^{(l)}$ denotes the normalized weight assigned to the $k$th non-self candidate.

Next, the preference-aligned positive representation is computed as the weighted combination of non-self candidate embeddings:

$$\mathbf{z}_u^{+(l)} = \sum_{\mathbf{e}_k^{pool}\in\mathcal{E}^{pool}\setminus\{\mathbf{e}_u^{(l)}\}} \alpha_{u,k}^{(l)} \cdot \mathbf{e}_k^{pool}, \quad \mathbf{z}_u^{+(l)} \in \mathbb{R}^d, \tag{18}$$

where $\mathbf{z}_u^{+(l)}$ denotes the positive proxy representation for user $u$ at layer $l$. This formulation performs a soft aggregation of semantically relevant candidates guided by the user's initial preferences, providing a more stable and informative positive signal than a single positive instance.

The preference-level contrastive loss is defined as:

$$\mathcal{L}_{\text{pref}}^{(l)}(u) = -\log\frac{\exp((\mathbf{e}_u^{(l)\top}\mathbf{z}_u^{+(l)})/\tau)}{\sum_{\mathbf{e}_k^{pool}\in\mathcal{E}^{pool}\setminus\{\mathbf{e}_u^{(l)}\}}\exp((\mathbf{e}_u^{(l)\top}\mathbf{e}_k^{pool})/\tau)}, \tag{19}$$

where $\mathbf{e}_u^{(l)} \in \mathbb{R}^d$ denotes the structural representation of user $u$ at layer $l$ after preference evolution attention aggregation and serves as the anchor in the preference-level contrastive objective. This objective maximizes the similarity between $\mathbf{e}_u^{(l)}$ and $\mathbf{z}_u^{+(l)}$ while contrasting it against the remaining non-self candidate embeddings in the current mini-batch. These remaining embeddings are treated as negative samples and are constructed from user representations across different propagation layers within the same mini-batch, excluding the current anchor.

The final preference-level loss is:

$$\mathcal{L}_{\text{pref}} = \sum_{l=1}^{L}\mathcal{L}_{\text{pref}}^{(l)}. \tag{20}$$

This loss aggregates the contrastive objective across all propagation layers, ensuring that structural representations remain aligned with the user's initial preferences throughout the diffusion process. Consequently, the model maintains global semantic consistency, mitigates semantic drift, and improves recommendation accuracy. This global alignment also provides the foundation for the subsequent path-level contrast, as detailed in Algorithm 3. The time complexity of the algorithm is $O(|\mathcal{U}|LNd)$, and the space complexity is $O(Nd)$, where $|\mathcal{U}|$ denotes the number of users, $L$ represents the number of propagation layers, $N$ is the size of the candidate embedding pool, and $d$ denotes the embedding dimension.

### 4.5. Path-level contrast

As illustrated in Fig. 2(E), the semantic path modeling module captures multi-hop relational dependencies in the KG. Rather than relying solely on structural proximity, it explicitly models the compositional semantics of multi-hop paths and integrates them with preference-aware

**Algorithm 3** Preference-level contrastive learning.

**Input:** Initial user preference vectors $\mathbf{p}_u^{(0)}$, User structural embeddings at layer $l$: $\mathbf{e}_u^{(l)}$, Candidate embedding pool $\{\mathbf{e}_k^{pool}\}_{k=1}^{N}$, Temperature parameter $\tau$, Layer number $L$.
**Output:** Preference-level contrastive loss $\mathcal{L}_{\text{pref}}$.

Initialize total loss $\mathcal{L}_{\text{pref}} = 0$;
// **Iterative Preference Alignment**
**for** each user $u$ **do**
  **for** layer $l = 1$ to $L$ **do**
    *// 1. Construct Preference-Aligned Positive Proxy*
    Exclude the current anchor from the candidate pool:
    $\widetilde{\mathcal{E}}_{u,l}^{pool} = \mathcal{E}^{pool} \setminus \{\mathbf{e}_u^{(l)}\}$;
    Compute similarity with candidate pool:
    $s_{u,k}^{(l)} = \mathbf{p}_u^{(0)\top} \mathbf{e}_k^{pool}, \quad \forall \mathbf{e}_k^{pool} \in \widetilde{\mathcal{E}}_{u,l}^{pool}$;
    Compute weighting distribution:
    $\alpha_{u,k}^{(l)} = \frac{\exp(s_{u,k}^{(l)}/\tau)}{\sum_{\mathbf{e}_m^{pool} \in \widetilde{\mathcal{E}}_{u,l}^{pool}} \exp(s_{u,m}^{(l)}/\tau)}$;
    Construct positive representation:
    $\mathbf{z}_u^{+(l)} = \sum_{\mathbf{e}_k^{pool} \in \widetilde{\mathcal{E}}_{u,l}^{pool}} \alpha_{u,k}^{(l)} \cdot \mathbf{e}_k^{pool}$;
    *// 2. Compute Layer-wise Contrastive Loss*
    Calculate contrastive loss:
    $\mathcal{L}_{\text{pref}}^{(l)}(u) = -\log \frac{\exp((\mathbf{e}_u^{(l)\top} \mathbf{z}_u^{+(l)})/\tau)}{\sum_{\mathbf{e}_k^{pool} \in \widetilde{\mathcal{E}}_{u,l}^{pool}} \exp((\mathbf{e}_u^{(l)\top} \mathbf{e}_k^{pool})/\tau)}$;
    *// 3. Accumulate Loss*
    $\mathcal{L}_{\text{pref}} = \mathcal{L}_{\text{pref}} + \mathcal{L}_{\text{pref}}^{(l)}(u)$;
  **end for**
**end for**
**return** $\mathcal{L}_{\text{pref}}$.

representations, yielding more expressive path-level representations for CL. For any user or item $x$ (where $x \in \{u, i\}$), we denote the set of triples at the $l$th layer as $\mathcal{T}_x^{(l)}$. Each element in this set represents a path sequence ending at layer $l$, allowing the model to incrementally accumulate entity and relation semantics along the propagation path.

Specifically, for a single path $p \in \mathcal{T}_x^{(l)}$, we aggregate the history of head entities and relations through summation and element-wise multiplication, respectively:

$$\mathbf{h}_p^{(l)} = \sum_{k=0}^{l} \mathbf{h}_p^{(k)}, \quad \mathbf{r}_p^{(l)} = \prod_{k=0}^{l} \mathbf{r}_p^{(k)}, \tag{21}$$

where $\mathbf{h}_p^{(k)}$ and $\mathbf{r}_p^{(k)}$ denote the embeddings of the entity and relation at the $k$th hop of path $p$. This operation captures both the structural trajectory and the compositional semantics of the path.

To construct the full path embedding for entity $x$ at layer $l$, the model aggregates these path representations using mean pooling:

$$\mathbf{Path}_x^{(l)} = \left[ \frac{1}{|\mathcal{T}_x^{(l)}|} \sum_{p \in \mathcal{T}_x^{(l)}} \mathbf{h}_p^{(l)} \,\|\, \frac{1}{|\mathcal{T}_x^{(l)}|} \sum_{p \in \mathcal{T}_x^{(l)}} \mathbf{r}_p^{(l)} \right], \tag{22}$$

where $\|$ denotes vector concatenation, and $\mathbf{Path}_x^{(l)} \in \mathbb{R}^{2D}$. Through this compositional modeling, the model learns hierarchical semantic path representations that serve as the foundation for subsequent path-level CL.

As shown in Fig. 2(F), the path-level CL module aligns direct paths derived from explicit interactions with indirect paths inferred from collaborative behaviors. Representations from every propagation layer contribute to the contrastive objective, encouraging semantic consistency throughout the multi-hop propagation process.

For each user $u$, let $\mathbf{Path}_u^{(l)}$ and $\mathbf{Path}_u^{(l)'}$ denote the direct-path and indirect-path embeddings at layer $l$. The model enforces semantic alignment through a bidirectional contrastive objective. First, both embeddings are normalized:

$$\mathbf{u}^{(l)} = \frac{\mathbf{Path}_u^{(l)}}{\|\mathbf{Path}_u^{(l)}\|_2}, \quad \mathbf{v}^{(l)} = \frac{\mathbf{Path}_u^{(l)'}}{\|\mathbf{Path}_u^{(l)'}\|_2}, \tag{23}$$

where $\mathbf{u}^{(l)}$ and $\mathbf{v}^{(l)}$ are the normalized direct-path and indirect-path representations for user $u$ at layer $l$.

The model then computes similarities in both directions. When the direct-path embedding $\mathbf{u}^{(l)}$ serves as the anchor, the positive similarity and the similarity with a negative sample $\mathbf{u}' \in \mathcal{U}'^{(l)}$ are computed as:

$$\text{sim}_u^{+} = \mathbf{u}^{(l)} \cdot \mathbf{v}^{(l)}, \qquad \text{sim}_{u,\mathbf{u}'}^{-} = \mathbf{u}^{(l)} \cdot \mathbf{u}', \tag{24}$$

where $\mathcal{U}'^{(l)}$ represents the set of direct-path embeddings from other users in the same mini-batch, which forms the negative pool.

Similarly, when $\mathbf{v}^{(l)}$ serves as the anchor:

$$\text{sim}_v^{+} = \mathbf{v}^{(l)} \cdot \mathbf{u}^{(l)}, \qquad \text{sim}_{v,\mathbf{v}'}^{-} = \mathbf{v}^{(l)} \cdot \mathbf{v}'. \tag{25}$$

The path-level InfoNCE loss is computed bidirectionally. When the direct-path embedding serves as the anchor, the loss is formulated as:

$$\mathcal{L}_u^{(l)} = -\log \frac{\exp(\text{sim}_u^{+}/\tau)}{\sum_{\mathbf{u}' \in \mathcal{U}'^{(l)}} \exp(\text{sim}_{u,\mathbf{u}'}^{-}/\tau)}. \tag{26}$$

Similarly, when the indirect-path embedding serves as the anchor, the loss is defined as:

$$\mathcal{L}_v^{(l)} = -\log \frac{\exp(\text{sim}_v^{+}/\tau)}{\sum_{\mathbf{v}' \in \mathcal{V}'^{(l)}} \exp(\text{sim}_{v,\mathbf{v}'}^{-}/\tau)}. \tag{27}$$

The path-level contrastive loss for user $u$ at layer $l$ is the sum of the losses from both directions:

$$\mathcal{L}_{\text{path}}^{(l)}(u) = \mathcal{L}_u^{(l)} + \mathcal{L}_v^{(l)}. \tag{28}$$

This bidirectional formulation encourages each direct-path embedding to remain close to its corresponding indirect-path embedding, and vice versa.

While the above derivation focuses on users, item-side path modeling follows the same procedure. Specifically, for each item $i$, we construct direct and indirect semantic paths and compute the corresponding item-side contrastive loss $\mathcal{L}_{\text{path}}^{(l)}(i)$ using the same bidirectional objective. The final path-level contrastive loss aggregates the losses from both users and items across all propagation layers:

$$\mathcal{L}_{\text{path}} = \sum_{l=1}^{L} \left( \mathcal{L}_{\text{path}}^{(l)}(u) + \mathcal{L}_{\text{path}}^{(l)}(i) \right). \tag{29}$$

This objective enforces multi-hop semantic consistency for both users and items by aligning direct and indirect path representations. As a result, the model captures richer latent semantic dependencies while complementing the preference-level contrastive objective through semantic alignment at the path level. The complete procedure is summarized in Algorithm 4. Together with the dual-path diffusion and preference evolution attention mechanisms, this framework effectively reduces the impact of noisy indirect preferences while preserving their complementary semantic information. The time complexity of the algorithm is $O\big((|\mathcal{U}| + |\mathcal{I}|)KL^2d + B^2Ld\big)$, and the space complexity is $O\big((|\mathcal{U}| + |\mathcal{I}|)Ld + B^2\big)$, where $|\mathcal{U}|$ and $|\mathcal{I}|$ denote the numbers of users and items, respectively, $L$ represents the number of propagation layers, $K$ denotes the number of sampled paths per layer, $B$ is the mini-batch size, and $d$ denotes the embedding dimension.

### 4.6. Model prediction and optimization

Finally, regardless of the chosen aggregation strategy, the comprehensive representation for each user and item is obtained by fusing the corresponding direct and indirect views. As illustrated in Fig. 2(G), we concatenate the aggregated direct embedding $\mathbf{e}_x^{\text{dir}}$ and indirect embedding $\mathbf{e}_x^{\text{ind}}$ to integrate both explicit interaction signals and implicit collaborative semantics:

$$\mathbf{e}_x = \mathbf{e}_x^{\text{dir}} \oplus \mathbf{e}_x^{\text{ind}}, \tag{30}$$

**Algorithm 4** Path-level contrastive learning.

**Input:** Triple sets $\mathcal{T}_x^{(l)}$ for users and items, Entity/Relation embeddings $\{\mathbf{h}, \mathbf{r}\}$, Temperature parameter $\tau$, Layer number $L$.
**Output:** Final path-level loss $\mathcal{L}_{\text{path}}$.

// **1. Semantic Path Modeling**
**for** each entity $x \in \{u, i\}$ **do**
  **for** $l = 1$ to $L$ **do**
    *For each single path $p \in \mathcal{T}_x^{(l)}$, accumulate history:*
    $\mathbf{h}_p^{(l)} = \sum_{k=0}^{l} \mathbf{h}_p^{(k)}, \quad \mathbf{r}_p^{(l)} = \prod_{k=0}^{l} \mathbf{r}_p^{(k)};$
    *Aggregate paths via mean pooling:*
    $\mathbf{Path}_x^{(l)} = \left[ \frac{1}{|\mathcal{T}_x^{(l)}|} \sum_p \mathbf{h}_p^{(l)} \parallel \frac{1}{|\mathcal{T}_x^{(l)}|} \sum_p \mathbf{r}_p^{(l)} \right];$
  **end for**
**end for**
// **2. Path-Level Contrast**
**for** each user $u$ **do**
  **for** $l = 1$ to $L$ **do**
    *Normalize embeddings:*
    $\mathbf{u}^{(l)} = \frac{\mathbf{Path}_u^{(l)}}{\|\mathbf{Path}_u^{(l)}\|_2}, \quad \mathbf{v}^{(l)} = \frac{\mathbf{Path}_u^{(l)'}}{\|\mathbf{Path}_u^{(l)'}\|_2};$
    *Compute Similarities:*
    $\text{sim}_u^{+} = \mathbf{u}^{(l)} \cdot \mathbf{v}^{(l)}, \quad \text{sim}_{u,\mathbf{u}'}^{-} = \mathbf{u}^{(l)} \cdot \mathbf{u}';$
    $\text{sim}_v^{+} = \mathbf{v}^{(l)} \cdot \mathbf{u}^{(l)}, \quad \text{sim}_{v,\mathbf{v}'}^{-} = \mathbf{v}^{(l)} \cdot \mathbf{v}';$
    *Compute InfoNCE Losses:*
    $\mathcal{L}_u^{(l)} = -\log \frac{\exp(\text{sim}_u^{+}/\tau)}{\sum_{\mathbf{u}' \in \mathcal{V}^{(l)}} \exp(\text{sim}_{u,\mathbf{u}'}^{-}/\tau)};$
    $\mathcal{L}_v^{(l)} = -\log \frac{\exp(\text{sim}_v^{+}/\tau)}{\sum_{\mathbf{v}' \in \mathcal{V}^{(l)}} \exp(\text{sim}_{v,\mathbf{v}'}^{-}/\tau)};$
    *Sum bidirectional loss:*
    $\mathcal{L}_{\text{path}}^{(l)}(u) = \mathcal{L}_u^{(l)} + \mathcal{L}_v^{(l)};$
  **end for**
**end for**
// **3. Item Path-Level Contrast**
Apply the same procedure to item entities to obtain $\mathcal{L}_{\text{path}}^{(l)}(i)$;
// **4. Final Aggregation**
Aggregate losses from users and items across layers:
$\mathcal{L}_{\text{path}} = \sum_{l=1}^{L} \left( \mathcal{L}_{\text{path}}^{(l)}(u) + \mathcal{L}_{\text{path}}^{(l)}(i) \right);$
**return** $\mathcal{L}_{\text{path}}$.

where $\oplus$ denotes vector concatenation. Consequently, user $u$ and item $i$ are represented by the final unified embeddings $\mathbf{e}_u, \mathbf{e}_i \in \mathbb{R}^d$.

Using these learned representations, the interaction score $\hat{y}_{ui}$ between user $u$ and item $i$ is predicted as their inner product:

$$\hat{y}_{ui} = \mathbf{e}_u^{\top} \mathbf{e}_i, \tag{31}$$

where a higher score indicates a stronger predicted user preference.

During training, the model jointly optimizes recommendation accuracy and semantic consistency. The overall loss function is defined as:

$$\mathcal{L} = \mathcal{L}_{rec} + \lambda_1 \mathcal{L}_{str} + \lambda_2 \mathcal{L}_{path} + \lambda_3 \|\Theta\|_2^2, \tag{32}$$

where $\mathcal{L}_{rec}$ denotes the recommendation loss, $\mathcal{L}_{str}$ and $\mathcal{L}_{path}$ represent the preference-level and path-level contrastive losses, respectively, and $\|\Theta\|_2^2$ is the $L_2$ regularization term used to prevent overfitting. The coefficients $\lambda_1$, $\lambda_2$, and $\lambda_3$ balance the different optimization objectives. In the implementation, $\lambda_1$ and $\lambda_2$ control the contributions of the preference-level and path-level contrastive losses, respectively.

The recommendation loss follows the Bayesian Personalized Ranking (BPR) objective, which encourages the model to assign higher scores to observed user-item interactions than to unobserved interactions:

$$\mathcal{L}_{rec} = -\sum_{(u,i,j) \in \mathcal{D}} \log \sigma(\hat{y}_{ui} - \hat{y}_{uj}), \tag{33}$$

where $(u, i, j)$ represents a training triplet consisting of a positive item $i$ and a negative item $j$, and $\sigma(\cdot)$ denotes the sigmoid function. The total loss $\mathcal{L}$ is minimized by updating all trainable parameters through backpropagation. This joint optimization enables the model to balance recommendation accuracy with representation consistency, resulting in more robust preference learning across both structural and semantic spaces.

## 5. Experiment

### 5.1. Experimental settings

All experiments were performed in Python 3.9.21 with the PyTorch 2.1.1 framework and CUDA 12.1 to ensure consistency and reproducibility.

To optimize the key hyperparameters of our model, we performed grid search over predefined candidate sets. Specifically, as shown in Fig. 3, the learning rate was selected from $\{2 \times 10^{-3}, 4 \times 10^{-3}, 6 \times 10^{-3}, 8 \times 10^{-3}, 1 \times 10^{-2}\}$, where $4 \times 10^{-3}$ achieved the best validation performance across all datasets. For the regularization coefficient, we searched within $\{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}, 10^{-7}\}$. For the contrastive learning-related coefficients, we explored values in $\{10^{-4}, 10^{-5}, 10^{-6}, 10^{-7}, 10^{-8}\}$. The final values were selected based on validation performance. For *Last.FM* and *Book-Crossing*, the optimal values for the L2 regularization coefficient, the preference-level contrast loss coefficient, and the path-level contrast loss coefficient were $10^{-4}$, $10^{-6}$, and $10^{-6}$, respectively, while for the denser *MovieLens-20M* dataset, smaller coefficients $10^{-6}$, $10^{-7}$, and $10^{-7}$ achieved better performance. This observation is consistent with the general intuition that denser datasets typically require weaker regularization, whereas sparser datasets benefit from relatively stronger regularization to alleviate overfitting.

For the embedding dimensions of entities and relations in the knowledge graph, we searched over $\{16, 32, 64, 128, 256\}$. For neighborhood sampling, the size of the triple set in each diffusion layer was also tuned within $\{16, 32, 64, 128, 256\}$ to balance computational efficiency and semantic coverage. The attention network was empirically designed as a two-layer structure, with hidden dimensions aligned with the embedding size.

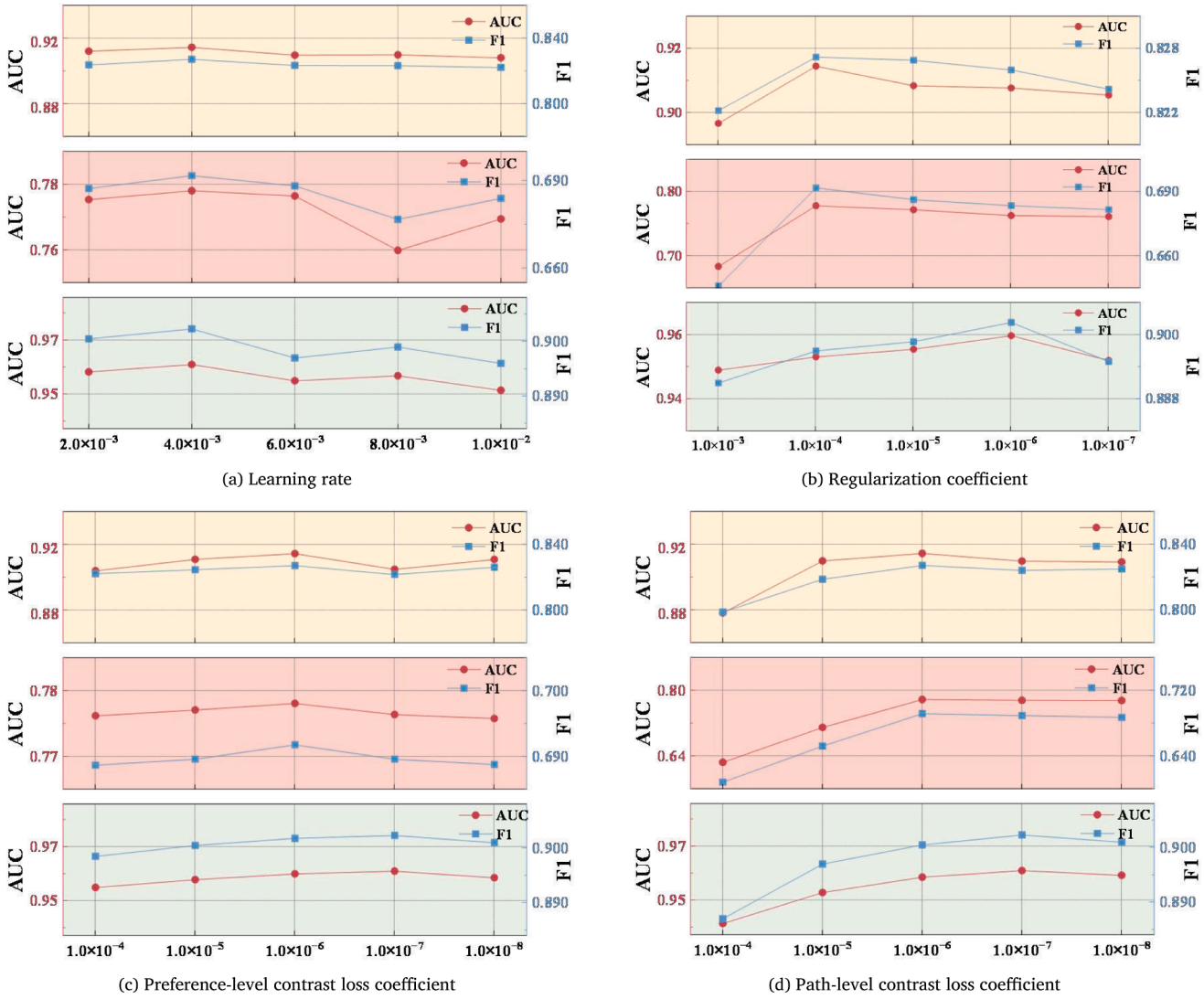
All baseline methods were implemented in the same environment, with hyperparameters either tuned within similar ranges or adopted from their original papers for fair comparison. We used the Adam optimizer (Kingma & Ba, 2014) with a batch size of 1024 and initialized all parameters using Xavier initialization (Glorot & Bengio, 2010).

### 5.2. Datasets

We evaluated the proposed model's effectiveness by conducting experiments on three benchmark datasets from diverse domains: **Movie**, **Book**, and **Music**. These datasets cover diverse recommendation scenarios, including movie watching, book reading, and music listening, thus providing a heterogeneous and comprehensive testbed for assessing both model accuracy and generalization ability.

- **Last.FM**[1] contains user listening histories, including artist play counts, from the Last.fm music platform. Since only listening frequencies are recorded, all interactions are considered implicit positive feedback. The knowledge graph associated with it has 9366 entities, 15,518 triples, and 60 relation types.
- **Book-Crossing**[2] collects user ratings for various books from the Book-Crossing community. Ratings above zero are treated as positive signals, while others are ignored. The related knowledge graph includes 77,903 entities, 151,500 triples, and 25 relation types.

[1] https://grouplens.org/datasets/hetrec-2011/.
[2] http://www2.informatik.uni-freiburg.de/cziegler/BX/.




(a) Learning rate

(b) Regularization coefficient

(c) Preference-level contrast loss coefficient

(d) Path-level contrast loss coefficient

**Fig. 3.** Parameter sensitivity analysis of DPRec under different hyperparameter settings. In each subfigure, the three panels from top to bottom correspond to the Music, Book, and Movie datasets, respectively. Red curves denote AUC, and blue curves denote F1.

- **MovieLens-20M**[3] consists of around 20 million user ratings, ranging from 1 to 5, for various movies on the MovieLens platform. Ratings of 4 or higher are considered positive feedback. The knowledge graph includes 102,569 entities, 499,474 triples, and 32 relation types.

During data preprocessing, all users and items were re-indexed into consecutive integer IDs. For each user, negative samples were generated by randomly selecting unobserved items in equal number to the positive ones, ensuring dataset balance. Each dataset's user–item interaction matrix was randomly split into 60% training, 20% validation, and 20% testing sets. The overall statistical details of these three datasets, including the numbers of users, items, interactions, and knowledge graph components, are summarized in Table 1.

### 5.3. Baselines

To thoroughly assess our model's effectiveness, we compare it with several representative baselines from various categories, including traditional collaborative filtering, knowledge graph-based methods, and graph neural network approaches. The following models are selected for comparison:

**Table 1**
Statistics of the three benchmark datasets employed in our experiments.

| Datasets | Music | Book | Movie |
|---|---|---|---|
| #users | 1872 | 17,860 | 98,741 |
| #items | 3864 | 14,910 | 16,954 |
| #interactions | 42,346 | 139,746 | 674,424 |
| #entities | 9366 | 77,903 | 102,569 |
| #relations | 60 | 25 | 32 |
| #triples | 15,518 | 151,500 | 499,474 |

- **BPRMF (2012)** (Rendle et al., 2012): A matrix factorization model optimized via BPR criterion, learning user and item latent representations by maximizing correct pairwise preferences from implicit feedback for personalized ranking.
- **CKE (2016)** (Zhang et al., 2016): A knowledge-aware recommendation model that simultaneously learns embeddings from collaborative filtering, item content, and knowledge graph structure to improve semantic representation.
- **RippleNet (2018)** (Wang et al., 2018): A memory-like propagation model that expands user preferences along knowledge graph links. It models multi-hop user interest diffusion to capture rich semantic associations between users and items.

[3] https://grouplens.org/datasets/movielens/.

- **KGCN (2019)** (Wang et al., 2019b): A graph convolutional network model that selectively aggregates information from entity neighbors in the knowledge graph, effectively capturing higher-order structural dependencies for recommendations.
- **KGNN-LS (2019)** (Wang et al., 2019a): An extension of KGCN that introduces label smoothing regularization and personalized weighting of knowledge graph edges, improving robustness and generalization in user–item embedding learning.
- **KGAT (2019)** (Wang et al., 2019c): A graph neural network-based recommendation model that fuses user-item interactions and knowledge graphs. It refines node embeddings via recursive attentive propagation of high-order relational information, enabling accurate and interpretable personalized recommendations.
- **CKAN (2020)** (Wang et al., 2020b): A collaborative knowledge-aware attention network that refines KGNN-LS by incorporating dual attention mechanisms to distinguish different types of relations and entity contributions.
- **KGIC (2022)** (Zou et al., 2022): A knowledge graph-based recommendation model that employs contrastive learning to align structural and semantic representations, thereby improving embedding consistency and recommendation accuracy.
- **KFGAN (2024)** (Wang et al., 2024): A fine-grained attention network with refined knowledge graph embedding that introduces entity-level attention and adaptive relational weighting to capture subtle semantic dependencies in user preferences.
- **PIFSA-GNN (2025)** (Li et al., 2025): A model that integrates *user preference interaction fusion* and *swap attention* mechanisms. It combines convolutional operations with a graph-based attention framework to balance entity importance, enhance semantic aggregation, and improve recommendation accuracy across heterogeneous domains.

### 5.4. Evaluation metrics

We use three common metrics to evaluate the model: AUC, F1-score, and Hit@K. These metrics measure the model's ability in both classification and ranking tasks.

**AUC (Area Under the ROC Curve)** serves as a standard metric for assessing classification models, with values spanning the interval $[0.5, 1]$, where higher scores signify superior performance. Mathematically, it is derived by calculating the area under the ROC curve, based on the ranking of predicted probabilities for positive versus negative instances.

**F1-score** is a metric that combines precision and recall, commonly used for imbalanced data. Precision indicates the proportion of correctly predicted positive items, while recall measures the model's ability to identify true positive items. They are defined as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}. \tag{34}$$

The F1-score combines the two and is defined as:

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{35}$$

A higher F1-score means that the model performs well in both precision and recall. It is a reliable measure when user–item interactions are sparse.

**Hit Ratio (Hit@K)** is one of the most commonly used ranking metrics in recommender systems. It evaluates whether the items that users actually interacted with appear in the top-$K$ recommendation results generated by the model. Specifically, if an item that a user clicked on or rated in the test set appears within the top-$K$ positions of the recommendation list, it is counted as a hit; otherwise, it is a miss. This metric intuitively reflects the coverage of the recommendation results. A higher Hit@K indicates that the model is more capable of including items that users are truly interested in within a limited-length recommendation list. In most experiments, Hit@1, Hit@5, and Hit@10 are reported to evaluate performance under different list lengths. In practical applications, Hit@K also reflects the usefulness of the recommender system in real-world front-end display scenarios.

**NDCG@K (Normalized Discounted Cumulative Gain)** is another widely used ranking metric in recommender systems. It evaluates not only whether the ground-truth item appears in the top-$K$ recommendation results, but also how well it is ranked. Specifically, NDCG introduces a logarithmic discount factor, assigning higher importance to items ranked at higher positions. As a result, NDCG ranges from 0 to 1, where a higher value indicates better ranking quality. This metric intuitively reflects the model's ability to place the most relevant items at the top of the recommendation list. In most experiments, NDCG@K is reported to evaluate ranking performance under different list lengths. In practical applications, NDCG is particularly important, as it emphasizes the correctness of top-ranked items, which directly affects user experience. The final NDCG scores are averaged over all users (1000 in this experiment), based on the ranking positions of the ground-truth items within the candidate sets. Since each user is evaluated with one sampled ground-truth item in the Top-K evaluation procedure, Hit@1 and NDCG@1 are equivalent. When $K = 1$, NDCG assigns a score of 1 only when the ground-truth item is ranked first, resulting in identical values for the two metrics in our experiments.

### 5.5. Performance comparison

#### 5.5.1. Evaluation based on AUC and F1

Table 2 presents the performance comparison of different models in terms of AUC and F1 across three benchmark datasets. All results are reported as the mean and standard deviation (in percentages) over multiple independent runs. Overall, DPRec consistently achieves the best performance across all datasets, demonstrating its effectiveness in both prediction accuracy and classification capability. On the Music dataset, DPRec outperforms all baseline methods, achieving improvements of 0.75% in AUC and 1.89% in F1 compared with the strongest baseline. The larger gain in F1 indicates that DPRec achieves a better balance between precision and recall, which is particularly important in practical recommendation scenarios. On the Book dataset, which is relatively sparse, DPRec demonstrates even greater improvements, with gains of 2.01% in AUC and 2.25% in F1. These results suggest that DPRec effectively captures user-item interactions while mitigating the adverse effects of data sparsity, resulting in more robust recommendation performance. On the Movie dataset, where user-item interactions are denser and most baseline models already achieve strong performance, DPRec continues to provide consistent improvements. Although the performance gains are smaller than those observed on the Music and Book datasets, they remain stable across repeated runs. This finding suggests that DPRec continues to refine user and item representations even when baseline performance approaches saturation. The larger improvements observed on sparse datasets can be attributed to the incomplete preference signals available in these settings. By jointly modeling complementary preference information and semantic relationships, DPRec is better able to recover latent preference patterns that are difficult for existing methods to capture. In contrast, denser datasets provide richer interaction information, allowing most baseline models to learn effective representations and leaving less room for improvement. Overall, these results demonstrate that DPRec consistently delivers superior classification performance across datasets with different sparsity characteristics, particularly in sparse recommendation scenarios while maintaining stable performance gains on denser datasets. In addition, statistical significance tests and effect size analyses based on five random seeds are provided in Table B.1.

#### 5.5.2. Evaluation based on Hit@K

Table 3 reports the performance comparison of different models in terms of Hit@K across three benchmark datasets. DPRec consistently achieves the best performance on all datasets and evaluation met-

**Table 2**
Performance comparison based on AUC and F1 scores across three benchmark datasets.

| Model | Year | Music | | Book | | Movie | |
|---|---|---|---|---|---|---|---|
| | | AUC(%) | F1(%) | AUC(%) | F1(%) | AUC(%) | F1(%) |
| BPRMF (Rendle et al., 2012) | 2012 | 86.55 ± 0.35 | 78.62 ± 0.55 | 65.11 ± 0.26 | 61.25 ± 0.56 | 89.22 ± 0.01 | 82.41 ± 0.02 |
| CKE (Zhang et al., 2016) | 2016 | 86.71 ± 0.50 | 78.70 ± 0.59 | 66.85 ± 1.12 | 62.46 ± 0.23 | 91.54 ± 0.88 | 84.38 ± 0.54 |
| RippleNet (Wang et al., 2018) | 2018 | 86.12 ± 0.52 | 78.12 ± 0.84 | 72.11 ± 0.58 | 64.72 ± 0.70 | 94.45 ± 0.03 | 87.86 ± 0.06 |
| KGCN (Wang et al., 2019b) | 2019 | 84.96 ± 2.94 | 77.79 ± 2.20 | 68.48 ± 1.31 | 63.45 ± 1.40 | 92.19 ± 0.70 | 85.01 ± 0.26 |
| KGNN-LS (Wang et al., 2019a) | 2019 | 85.75 ± 0.38 | 78.11 ± 0.44 | 68.53 ± 0.23 | 63.64 ± 0.38 | 92.37 ± 0.47 | 85.23 ± 0.64 |
| KGAT (Wang et al., 2019c) | 2019 | 86.59 ± 1.69 | 78.98 ± 1.76 | 72.02 ± 1.83 | 63.98 ± 2.09 | 93.01 ± 0.02 | 86.11 ± 0.03 |
| CKAN (Wang et al., 2020b) | 2020 | 87.01 ± 0.76 | 79.25 ± 0.85 | 74.36 ± 0.45 | 66.71 ± 0.48 | 94.26 ± 0.05 | 87.70 ± 0.11 |
| KGIC (Zou et al., 2022) | 2022 | 90.02 ± 0.22 | 81.11 ± 0.21 | 75.90 ± 0.04 | 67.51 ± 0.46 | 94.53 ± 0.17 | 87.92 ± 0.05 |
| KFGAN (Wang et al., 2024) | 2024 | 90.10 ± 0.31 | 81.19 ± 0.24 | 75.70 ± 0.16 | 67.20 ± 0.06 | 94.86 ± 0.11 | 88.12 ± 0.06 |
| PIFSA-GNN (Li et al., 2025) | 2025 | 90.85 ± 0.31 | 80.95 ± 0.67 | 76.29 ± 1.36 | 67.66 ± 0.76 | 95.68 ± 0.05 | 90.03 ± 0.03 |
| **DPRec** | | **91.53**[a] ± 0.28 | **82.72**[a] ± 0.34 | **77.82**[a] ± 0.23 | **69.18**[a] ± 0.18 | **95.97**[a] ± 0.04 | **90.23**[a] ± 0.07 |
| **Gain (%)** | | 0.75% | 1.89% | 2.01% | 2.25% | 0.30% | 0.22% |

[a] indicates statistical significance according to the $t$-test ($p < 0.05$).

**Table 3**
Comparison of Hit@K results on three benchmark datasets.

| Model | Music | | | Book | | | Movie | | |
|---|---|---|---|---|---|---|---|---|---|
| | Hit@1(%) | Hit@5(%) | Hit@10(%) | Hit@1(%) | Hit@5(%) | Hit@10(%) | Hit@1(%) | Hit@5(%) | Hit@10(%) |
| BPRMF | 23.40 ± 2.78 | 60.36 ± 1.96 | 70.30 ± 2.13 | 17.50 ± 0.87 | 42.76 ± 0.74 | 54.98 ± 0.43 | 19.02 ± 1.57 | 61.10 ± 0.91 | 80.86 ± 0.83 |
| CKE | 30.26 ± 3.50 | 59.68 ± 2.46 | 71.92 ± 2.32 | 16.20 ± 2.16 | 41.10 ± 1.89 | 53.32 ± 2.55 | 32.60 ± 1.22 | 80.52 ± 1.40 | 91.50 ± 1.16 |
| RippleNet | 25.34 ± 1.70 | 55.52 ± 2.24 | 69.38 ± 2.97 | 23.16 ± 1.25 | 46.04 ± 1.27 | 58.18 ± 1.12 | 24.22 ± 1.70 | 71.86 ± 1.52 | 85.68 ± 0.70 |
| KGCN | 29.10 ± 2.74 | 57.32 ± 1.50 | 67.84 ± 1.86 | 23.72 ± 0.50 | 48.68 ± 0.42 | 60.50 ± 1.29 | 33.90 ± 0.92 | 85.92 ± 1.98 | 91.40 ± 0.20 |
| KGNN-LS | 25.52 ± 0.76 | 54.40 ± 1.39 | 68.62 ± 0.35 | 23.86 ± 0.54 | 49.50 ± 0.60 | 61.58 ± 1.32 | 30.22 ± 0.36 | 71.46 ± 1.31 | 83.62 ± 0.15 |
| KGAT | 26.22 ± 4.30 | 57.42 ± 2.49 | 70.78 ± 2.02 | 24.00 ± 2.79 | 48.96 ± 1.61 | 60.54 ± 0.88 | 45.10 ± 0.78 | 86.32 ± 0.56 | 93.54 ± 0.63 |
| CKAN | 22.76 ± 1.06 | 58.80 ± 1.70 | 76.15 ± 1.63 | 24.60 ± 0.15 | 49.18 ± 1.77 | 60.60 ± 2.05 | 42.50 ± 1.56 | 84.86 ± 0.78 | 93.32 ± 0.65 |
| KGIC | 33.88 ± 0.78 | 67.52 ± 0.62 | 82.14 ± 0.82 | 25.89 ± 0.64 | 50.18 ± 1.35 | 61.30 ± 1.97 | 45.90 ± 0.91 | 87.40 ± 0.77 | 94.10 ± 0.65 |
| KFGAN | 35.10 ± 0.96 | 67.56 ± 0.76 | 80.90 ± 0.64 | 26.14 ± 0.32 | 50.34 ± 1.65 | 63.86 ± 1.21 | 46.12 ± 1.34 | 86.10 ± 1.14 | 94.24 ± 0.92 |
| PIFSA-GNN | 38.34 ± 1.19 | 68.06 ± 1.21 | 81.90 ± 1.40 | 25.94 ± 0.38 | 52.00 ± 0.76 | 63.90 ± 0.85 | 47.03 ± 0.14 | 87.13 ± 0.64 | 96.47 ± 0.49 |
| **DPRec** | **43.74**[a] ± 1.21 | **73.60**[a] ± 1.83 | **83.20**[a] ± 0.83 | **26.88**[a] ± 0.33 | **52.80**[a] ± 0.54 | **64.62**[a] ± 0.51 | **48.27**[a] ± 0.83 | **87.83**[a] ± 0.52 | **96.53** ± 0.10 |
| **Gain (%)** | 14.08% | 8.14% | 1.29% | 2.83% | 1.54% | 1.13% | 2.64% | 0.80% | 0.06% |

[a] indicates statistical significance according to the $t$-test ($p < 0.05$).

rics, demonstrating its effectiveness in capturing user preferences and improving recommendation accuracy. Moreover, the relatively small standard deviations across multiple independent runs indicate that the proposed model achieves stable performance with low variability. On the Music dataset, DPRec substantially outperforms all baseline methods, achieving improvements of 14.08%, 8.14%, and 1.29% on Hit@1, Hit@5, and Hit@10, respectively, compared with the strongest baseline, PIFSA-GNN. The particularly large improvement on Hit@1 demonstrates DPRec's superior ability to rank the most relevant items at the top of the recommendation list, which is especially important in real-world recommendation scenarios. The consistently low standard deviations further confirm the robustness of the proposed model. On the Book dataset, despite the overall lower performance caused by greater data sparsity, DPRec continues to outperform the strongest baseline, with improvements of 2.83%, 1.54%, and 1.13% on Hit@1, Hit@5, and Hit@10, respectively. The stable performance across repeated runs further demonstrates the model's ability to alleviate the effects of data sparsity while capturing meaningful user-item interactions. On the Movie dataset, where user-item interactions are relatively dense and most models already achieve strong performance, DPRec continues to deliver competitive results. Although DPRec achieves improvements on Hit@1, Hit@5, and Hit@10, the improvement on Hit@10 is comparatively small and does not reach statistical significance. This difference is primarily attributable to the characteristics of the Hit@10 metric, which only evaluates whether the ground-truth item appears within the top-10 recommendation list. As baseline models already exhibit strong retrieval capability on this dataset, there is relatively little room for further improvement. In contrast, DPRec primarily enhances fine-grained preference modeling and item ranking through dual-path preference diffusion and semantic alignment mechanisms, leading to more noticeable improvements on Hit@1. Since many candidate items in the Movie dataset exhibit similar interaction patterns, DPRec mainly refines the relative ranking of candidate items rather than substantially changing the overall Top-K recommendation set, resulting in limited improvements on Hit@5 and Hit@10. Overall, these results demonstrate that DPRec consistently improves recommendation performance across datasets with different sparsity characteristics. The model achieves statistically significant improvements on most Hit@K metrics while maintaining stable performance across multiple independent runs.

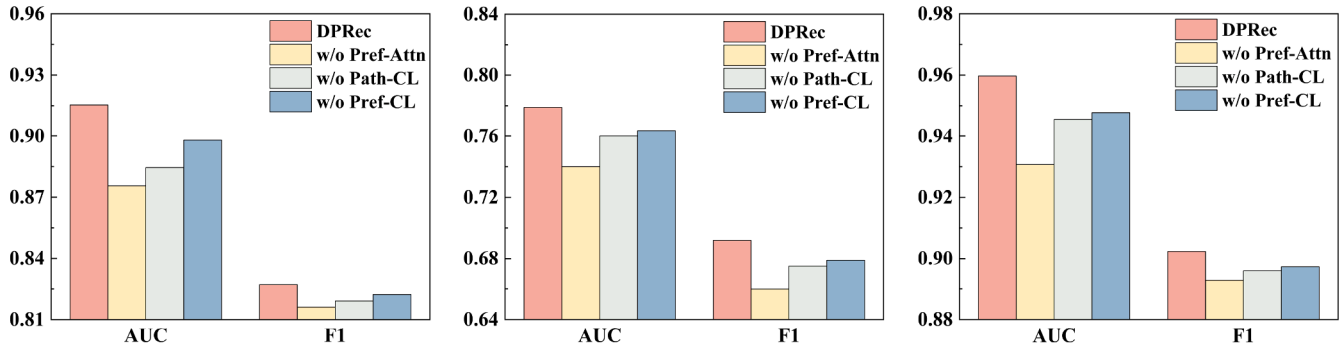
#### 5.5.3. Evaluation based on NDCG@K

Table 4 reports the performance comparison of different methods in terms of NDCG@K across three benchmark datasets. Overall, DPRec achieves the best performance on all datasets and evaluation metrics, further demonstrating its effectiveness in modeling ranking quality and capturing fine-grained user preferences. On the Music dataset, DPRec substantially outperforms all baseline methods. Compared with the strongest baseline, it improves NDCG@1, NDCG@5, and NDCG@10 by 14.08%, 5.89%, and 5.52%, respectively. The particularly large improvement in NDCG@1 indicates that DPRec is especially effective at ranking the most relevant items at the top of the recommendation list, consistent with its strong performance on the Hit@K metrics. On the Book dataset, where user-item interactions are relatively sparse, DPRec continues to outperform the strongest baseline across all evaluation metrics, achieving gains of 2.83%, 1.94%, and 1.49% on NDCG@1, NDCG@5, and NDCG@10, respectively. These results demonstrate that DPRec learns robust user and item representations even under sparse interaction conditions, resulting in improved ranking quality. On the

**Table 4**
Comparison of NDCG@K results on three benchmark datasets.

| Model | Music | | | Book | | | Movie | | |
|---|---|---|---|---|---|---|---|---|---|
| | NDCG@1(%) | NDCG@5(%) | NDCG@10(%) | NDCG@1(%) | NDCG@5(%) | NDCG@10(%) | NDCG@1(%) | NDCG@5(%) | NDCG@10(%) |
| BPRMF | 23.40 ± 2.78 | 53.87 ± 0.83 | 56.46 ± 0.67 | 17.50 ± 0.87 | 28.70 ± 0.16 | 32.68 ± 0.11 | 19.02 ± 1.57 | 63.60 ± 0.96 | 65.20 ± 0.88 |
| CKE | 30.26 ± 3.50 | 53.96 ± 1.42 | 58.65 ± 1.76 | 16.20 ± 2.16 | 26.12 ± 1.56 | 30.59 ± 0.98 | 32.60 ± 1.22 | 64.10 ± 1.28 | 66.82 ± 1.20 |
| RippleNet | 25.34 ± 1.70 | 54.54 ± 1.30 | 57.63 ± 1.10 | 23.16 ± 1.25 | 31.86 ± 0.16 | 35.18 ± 0.20 | 24.22 ± 1.70 | 62.21 ± 1.25 | 64.10 ± 0.82 |
| KGCN | 29.10 ± 2.74 | 54.58 ± 1.34 | 57.89 ± 1.77 | 23.72 ± 0.50 | 30.58 ± 3.46 | 34.13 ± 2.09 | 33.90 ± 0.92 | 62.13 ± 0.75 | 65.02 ± 1.14 |
| KGNN-LS | 25.52 ± 0.76 | 53.06 ± 0.89 | 56.89 ± 1.27 | 23.86 ± 0.54 | 31.36 ± 2.02 | 34.92 ± 1.39 | 30.22 ± 0.36 | 57.73 ± 0.47 | 61.18 ± 0.14 |
| KGAT | 26.22 ± 4.30 | 53.97 ± 1.20 | 57.93 ± 0.89 | 24.00 ± 2.79 | 30.98 ± 0.64 | 35.25 ± 0.71 | 45.10 ± 0.78 | 66.69 ± 1.45 | 69.16 ± 1.26 |
| CKAN | 22.76 ± 1.06 | 38.12 ± 2.96 | 45.79 ± 1.24 | 24.60 ± 0.15 | 31.65 ± 0.57 | 36.94 ± 0.34 | 42.50 ± 1.56 | 64.58 ± 0.77 | 70.23 ± 0.65 |
| KGIC | 33.88 ± 0.78 | 56.49 ± 0.19 | 59.62 ± 0.10 | 25.89 ± 0.64 | 34.42 ± 1.04 | 38.85 ± 1.08 | 45.90 ± 0.91 | 62.40 ± 0.92 | 68.84 ± 0.64 |
| KFGAN | 35.10 ± 0.96 | 54.57 ± 0.32 | 57.22 ± 0.28 | 26.14 ± 0.32 | 35.82 ± 1.26 | 39.24 ± 1.32 | 46.12 ± 1.34 | 63.70 ± 1.34 | 69.80 ± 0.78 |
| PIFSA-GNN | 38.34 ± 1.19 | 52.09 ± 0.55 | 56.87 ± 0.52 | 25.94 ± 0.38 | 38.59 ± 1.06 | 42.18 ± 0.32 | 47.03 ± 0.14 | 67.93 ± 0.75 | 71.13 ± 0.65 |
| **DPRec** | **43.74**[a] ± 1.21 | **59.82**[a] ± 1.34 | **62.91**[a] ± 1.03 | **26.88**[a] ± 0.33 | **39.34**[a] ± 0.57 | **42.81**[a] ± 0.49 | **48.27**[a] ± 0.83 | **70.57**[aa] ± 1.15 | **72.21**[*] ± 0.98 |
| **Gain (%)** | 14.08% | 5.89% | 5.52% | 2.83% | 1.94% | 1.49% | 2.64% | 3.89% | 1.52% |

[a] indicates statistical significance according to the $t$-test ($p < 0.05$).




(a) Music (b) Book (c) Movie

**Fig. 4.** Ablation study results on three datasets showing AUC and F1 variations after removing different modules from DPRec.

Movie dataset, where user-item interactions are denser and baseline models already achieve strong performance, DPRec continues to provide consistent improvements. Although the performance gains are smaller than those observed on the Music and Book datasets, they demonstrate that DPRec can further refine learned representations and improve the ranking of the most relevant items even when baseline performance approaches saturation. Overall, these results demonstrate that DPRec not only improves recommendation accuracy but also consistently enhances ranking quality, particularly for top-ranked recommendations, across datasets with different sparsity characteristics.

### 5.6. Model analysis

#### 5.6.1. Effect of preference evolution attention aggregation and contrastive learning modules

To comprehensively evaluate the contribution of each component, we conduct ablation experiments on the *Music*, *Book*, and *Movie* datasets by sequentially removing the Preference Evolution Attention Aggregation (Pref-Attn), Path-Level Contrast (Path-CL), and Preference-Level Contrast (Pref-CL) modules, as illustrated in Fig. 4a–c. Overall, the complete DPRec model consistently achieves the best performance across all datasets, demonstrating that these modules provide complementary benefits for preference representation learning and semantic alignment.

The Pref-Attn module is designed to adaptively capture the importance of neighboring entities during knowledge propagation. Removing this module replaces the adaptive attention mechanism with a standard mean aggregation strategy. On the *Music* dataset, DPRec achieves an AUC of 0.9153 and an F1 score of 0.827, whereas removing Pref-Attn reduces the AUC and F1 scores to 0.874 and 0.818, respectively. Similar performance degradation is observed on the *Book* and *Movie* datasets. The substantial performance drop indicates that the attention mechanism effectively identifies user-specific semantic information while suppressing irrelevant signals during propagation.

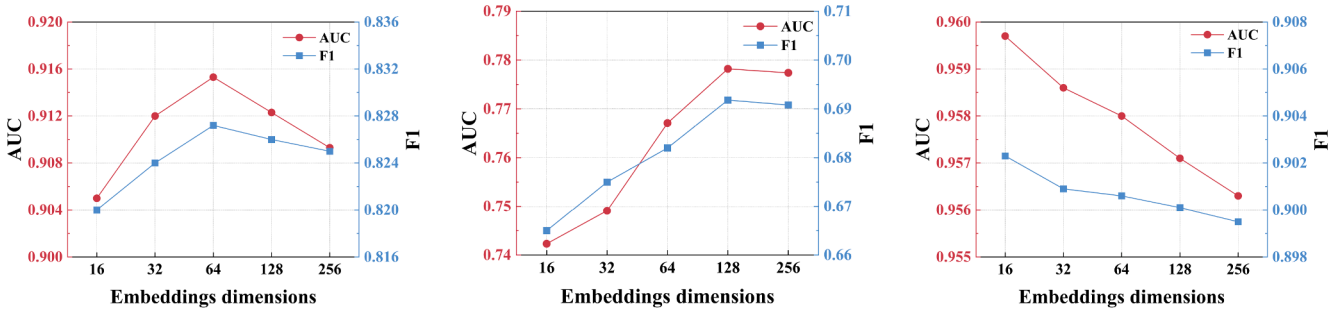
The Path-CL and Pref-CL modules provide complementary supervision from the perspectives of semantic paths and preference representations. Path-CL improves the consistency of multi-hop representations by exploiting complementary propagation paths (Li et al., 2026b), whereas Pref-CL enhances preference discrimination through preference-level semantic alignment. On the *Music* dataset, removing Path-CL reduces the AUC and F1 scores to 0.885 and 0.819, respectively, while removing Pref-CL results in AUC and F1 scores of 0.898 and 0.822. Similar trends are observed on the *Book* and *Movie* datasets.

Overall, the ablation study demonstrates that each component contributes to the overall effectiveness of DPRec. Pref-Attn enhances adaptive semantic aggregation by assigning appropriate weights to neighboring entities, Path-CL improves the consistency of semantic path representations, and Pref-CL strengthens preference-level semantic alignment. Together, these modules enable DPRec to learn more expressive user and item representations, resulting in more effective preference modeling and semantic alignment.

#### 5.6.2. Effect of embedding dimension

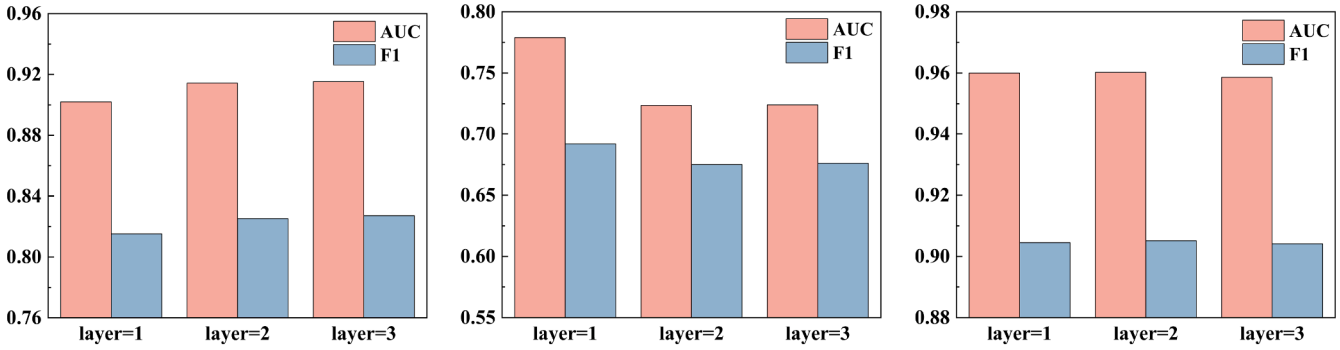
The embedding dimension $d$ controls the representational capacity of DPRec and directly influences its ability to capture semantic and preference information. We evaluate embedding dimensions of $\{16, 32, 64, 128, 256\}$ while keeping all other parameters fixed, as illustrated in Fig. 5a–5c.

For the *Music* dataset shown in Fig. 5a, performance initially improves as the embedding dimension increases and reaches its optimum at $d = 64$, where the AUC is approximately 0.915 and the F1 score is




(a) Music (b) Book (c) Movie

**Fig. 5.** Performance comparison under different embedding dimensions on three datasets.




(a) Music (b) Book (c) Movie

**Fig. 6.** Performance comparison under different propagation layers on three datasets.

about 0.827. Beyond this point, performance gradually declines, suggesting that a moderate embedding dimension provides an effective balance between representational capacity and generalization.

In contrast, the *Book* dataset shown in Fig. 5b benefits from larger embedding dimensions. Both AUC and F1 improve steadily as $d$ increases, with the AUC rising from approximately 0.742 at $d = 16$ to a peak of 0.778 at $d = 128$. These results suggest that richer embedding representations are beneficial for modeling the complex semantic relationships present in this dataset.

Conversely, the *Movie* dataset shown in Fig. 5c exhibits a monotonic decline in performance as $d$ increases. The best results are achieved with the smallest embedding dimension, $d = 16$, where the AUC is below 0.960 and the F1 score is about 0.902. This finding suggests that larger embedding dimensions may introduce unnecessary model complexity and increase the risk of overfitting for this dataset.

Overall, the optimal embedding dimension depends on the characteristics of the dataset. While the *Movie* dataset achieves the best performance with a relatively small embedding dimension, the *Music* and *Book* datasets benefit from larger embedding spaces, indicating that the appropriate model capacity should be selected according to the complexity of the underlying recommendation task.

### 5.6.3. *Effect of propagation layer depth*

This section investigates the impact of propagation depth on DPRec by varying the number of propagation layers from one to three while keeping all other settings unchanged. Fig. 6a–c presents the AUC and F1 results under different propagation depths on the *Music*, *Book*, and *Movie* datasets.

On the *Music* dataset, increasing the propagation depth from one to three layers improves the AUC from 0.910 to 0.915 and the F1 score from 0.813 to 0.827. This improvement indicates that deeper propagation enables DPRec to capture richer high-order collaborative information and more comprehensive semantic dependencies. Both two-layer and three-layer configurations achieve stable performance, while the three-layer model provides a slight improvement by incorporating additional contextual information.

On the *Book* dataset, the best performance is achieved with a single propagation layer. Further increasing the propagation depth leads to performance degradation, which can be attributed to the sparse interaction structure of this dataset. In such scenarios, excessive propagation may introduce irrelevant semantic information and accumulate noisy signals during knowledge aggregation, thereby weakening the quality of learned representations.

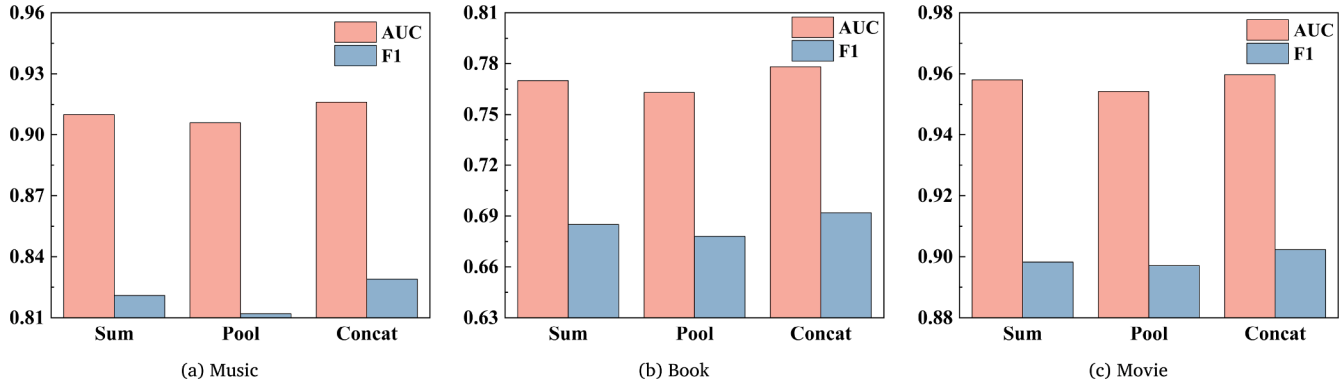
On the *Movie* dataset, the performance remains relatively stable across different propagation depths, with the AUC maintained around 0.960 and the F1 score around 0.902. This phenomenon suggests that the dense interaction structure of the dataset already provides sufficient collaborative signals for effective representation learning. Therefore, increasing the propagation depth beyond this point provides limited additional benefits while introducing extra computational costs.

Overall, the optimal propagation depth depends on the characteristics of different datasets. A shallow propagation strategy is more suitable for sparse datasets such as *Book*, where excessive information aggregation may amplify noise. In contrast, datasets with richer interaction information, such as *Music* and *Movie*, can benefit from deeper propagation to capture additional high-order semantic information. These results demonstrate that DPRec can effectively adapt its propagation depth to different recommendation scenarios.

### 5.6.4. *Effect of different aggregation strategies*

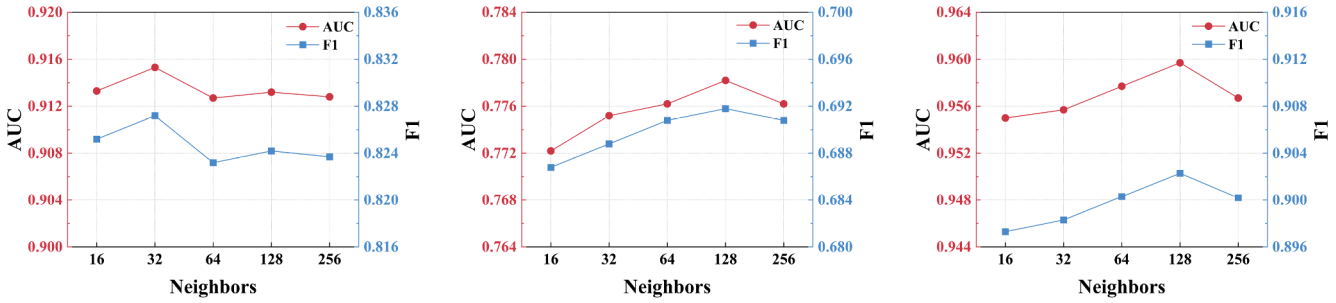
This section evaluates three aggregation strategies in DPRec, including DPRec-Sum, DPRec-Pool, and DPRec-Concat, under the same experimental settings on the *Music*, *Book*, and *Movie* datasets, as illustrated in Fig. 7a–c.

For the *Music* dataset shown in Fig. 7a, all three strategies achieve competitive performance, while DPRec-Concat obtains the best results,




(a) Music (b) Book (c) Movie

**Fig. 7.** Performance comparison of three aggregation strategies (Sum, Pool, Concat) on different datasets.




(a) Music (b) Book (c) Movie

**Fig. 8.** Effect of different neighbor sampling sizes on AUC and F1 performance across three datasets.

followed by DPRec-Sum and DPRec-Pool. This indicates that concatenating representations from different propagation layers can preserve more detailed preference information, which is beneficial for capturing diverse user interests.

For the *Book* dataset shown in Fig. 7b, DPRec-Concat achieves the highest performance, while DPRec-Sum and DPRec-Pool show relatively lower results. This suggests that retaining multi-level semantic information is particularly important in sparse scenarios, whereas pooling may lose informative features during aggregation.

For the *Movie* dataset shown in Fig. 7c, the performance differences among different strategies are relatively small. DPRec-Concat achieves the best performance with an AUC of approximately 0.960 and an F1 score of 0.902, while DPRec-Sum obtains an AUC of 0.958. This result indicates that dense interaction information reduces the impact of aggregation strategies, although concatenation still maintains a slight advantage.

Overall, DPRec-Concat consistently provides the best performance by preserving complementary information from different propagation layers, while DPRec-Sum achieves stable results with slightly lower performance. In contrast, DPRec-Pool may lose fine-grained semantic information due to feature compression.

### 5.6.5. *Effect of neighbor sampling size*

To study the influence of the neighbor sampling size, we vary the number of sampled neighbors across 16, 32, 64, 128, and 256 while keeping all other hyperparameters fixed. The results on the *Music*, *Book*, and *Movie* datasets are shown in Fig. 8a–c under identical experimental settings.

On the *Music* dataset, the best performance is achieved with 32 sampled neighbors. Compared with 16 neighbors, both the AUC and F1 score increase, whereas performance gradually declines as the neighborhood size grows beyond 32. This indicates that a moderate neighborhood size can effectively integrate semantic information without excessively averaging diverse preference signals.

Furthermore, on the *Book* dataset, performance improves steadily and reaches its peak at 128 neighbors, with an AUC of 0.778 and an F1 score of 0.692. A slight decline at 256 neighbors suggests that incorporating additional neighbors may introduce irrelevant semantic information in this sparse graph, thereby affecting the quality of learned representations.

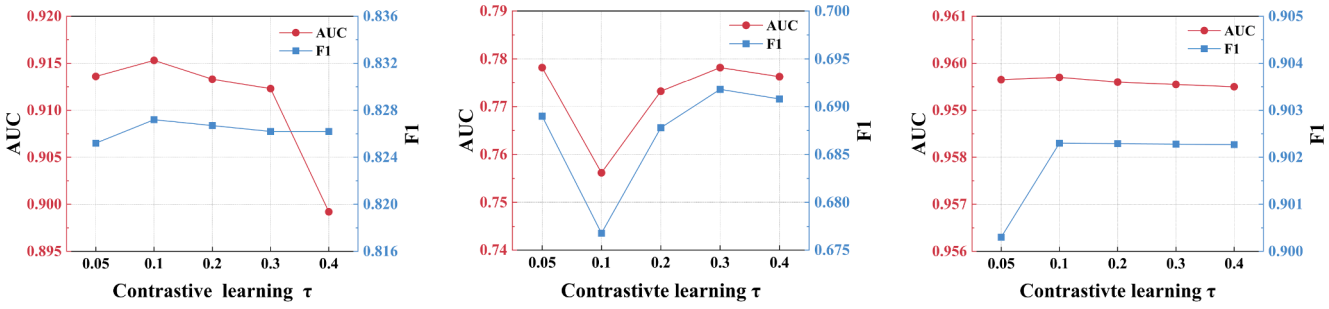
Finally, on the *Movie* dataset, performance also improves as the neighborhood size increases from 16 to 128. The AUC rises from 0.956 to 0.960, and the F1 score reaches 0.902, indicating that richer contextual information strengthens user-item matching. However, increasing the neighborhood size to 256 leads to a slight decline, which can be attributed to over-smoothing caused by excessive aggregation.

Overall, too few neighbors result in insufficient semantic aggregation, whereas too many introduce noisy or redundant information. The optimal sampling size varies across datasets: the *Book* and *Movie* datasets achieve the best performance with a neighborhood size of 128, whereas the *Music* dataset performs best with a smaller neighborhood size of 32. This indicates that the appropriate sampling size should be determined according to the characteristics of different recommendation scenarios.

### 5.6.6. *Effect of temperature coefficient $\tau$*

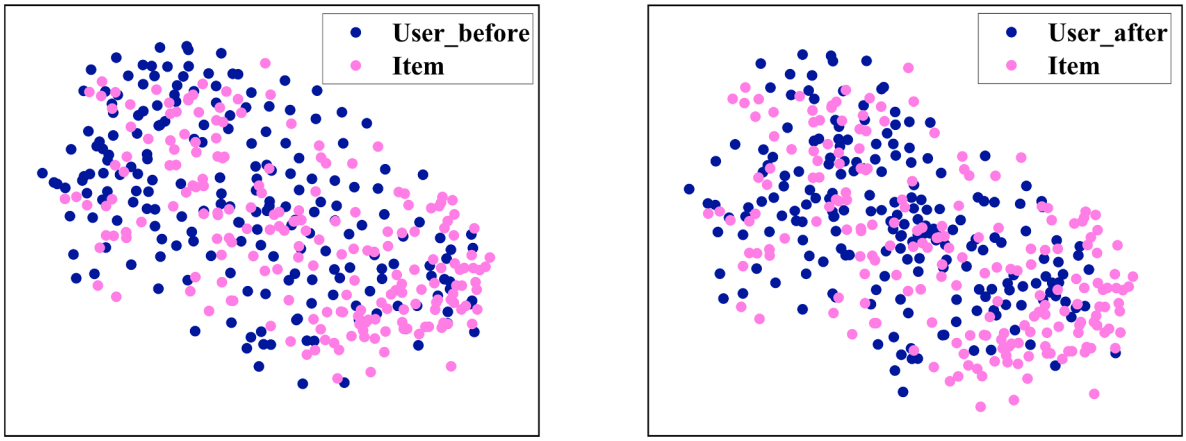
This section investigates the effect of the temperature coefficient $\tau$, which controls the sharpness of the InfoNCE similarity distribution. A smaller $\tau$ emphasizes hard positive and negative samples, whereas a larger $\tau$ produces a smoother contrastive objective. We vary $\tau$ over the set $\{0.05, 0.1, 0.2, 0.3, 0.4\}$ and report the results on the *Music*, *Book*, and *Movie* datasets in Fig. 9a–c.

On the *Music* dataset, the best performance is achieved at $\tau = 0.1$, where both the AUC ($\approx 0.915$) and F1 score ($\approx 0.827$) reach their highest values. Deviating from this value causes performance to decline, indicating that $\tau = 0.1$ provides an effective balance between emphasizing hard negatives and maintaining training stability.




(a) Music (b) Book (c) Movie

**Fig. 9.** Effect of temperature coefficient $\tau$ on AUC and F1 across datasets.




(a) Before training (b) After training

**Fig. 10.** Visualization of user and item embeddings before and after training.

In contrast, the *Book* dataset exhibits a fluctuating trend. Performance decreases noticeably at $\tau = 0.1$ but recovers to achieve the best results at $\tau = 0.3$ (AUC $\approx$ 0.778, F1 $\approx$ 0.692). This suggests that, for this sparse dataset, a moderate temperature better balances intra-class compactness and inter-class separation.

For the *Movie* dataset, the results are relatively stable. Performance improves substantially from $\tau = 0.05$ to $\tau = 0.1$ and then remains at a consistently high level, with the best trade-off achieved at $\tau = 0.1$ (AUC $\approx$ 0.9597). This indicates that the dense user-item interactions in the *Movie* dataset make the model less sensitive to larger $\tau$ values, provided that $\tau$ is not excessively small.

Overall, the optimal temperature coefficient varies across datasets. The *Music* and *Movie* datasets achieve their best performance at $\tau = 0.1$, whereas the *Book* dataset performs best at $\tau = 0.3$, indicating that the appropriate temperature should be selected according to the characteristics of the dataset.

#### 5.6.7. Analysis of potential false negatives in-batch contrastive learning

Although in-batch CL efficiently constructs negative samples, it may introduce potential false negatives because samples within the same mini-batch can share similar preferences or semantic characteristics. To investigate this issue, we estimate the proportion of potential false negatives based on interaction overlap. Specifically, the Jaccard similarity between users is calculated according to their interaction sets:

$$J(u_i, u_j) = \frac{|I_{u_i} \cap I_{u_j}|}{|I_{u_i} \cup I_{u_j}|}, \tag{36}$$

where $I_u$ denotes the positive interaction set of user $u$. We evaluate potential false-negative cases under different Jaccard thresholds (0.0, 0.1, and 0.2), where larger thresholds indicate stricter criteria for identifying semantically similar negative samples. A pair with a Jaccard similarity greater than the corresponding threshold is considered as a potential false-negative candidate. The analysis is conducted using a batch size of 1024 and five randomly constructed mini-batches. The results are summarized in Table 5. Although a small proportion of the in-batch negative samples may exhibit potential semantic correlations with the anchor, the vast majority of negative samples remain valid. These results indicate that the false-negative issue exists but is relatively limited in the evaluated datasets.

**Table 5**
Detailed potential false-negative analysis under different Jaccard thresholds.

| Dataset | Threshold | User-side (%) | Item-side (%) |
|---|---|---|---|
| Music | 0.0 | 11.13 | 5.10 |
| | 0.1 | 2.14 | 0.25 |
| | 0.2 | 0.29 | 0.09 |
| Book | 0.0 | 15.71 | 5.84 |
| | 0.1 | 0.50 | 0.33 |
| | 0.2 | 0.24 | 0.14 |
| Movie | 0.0 | 2.76 | 13.70 |
| | 0.1 | 0.91 | 0.05 |
| | 0.2 | 0.19 | 0.05 |

### 5.7. Visualization

#### 5.7.1. Embedding visualization analysis

Representation learning maps users and items into high-dimensional latent vectors to capture underlying semantic and preference informa-

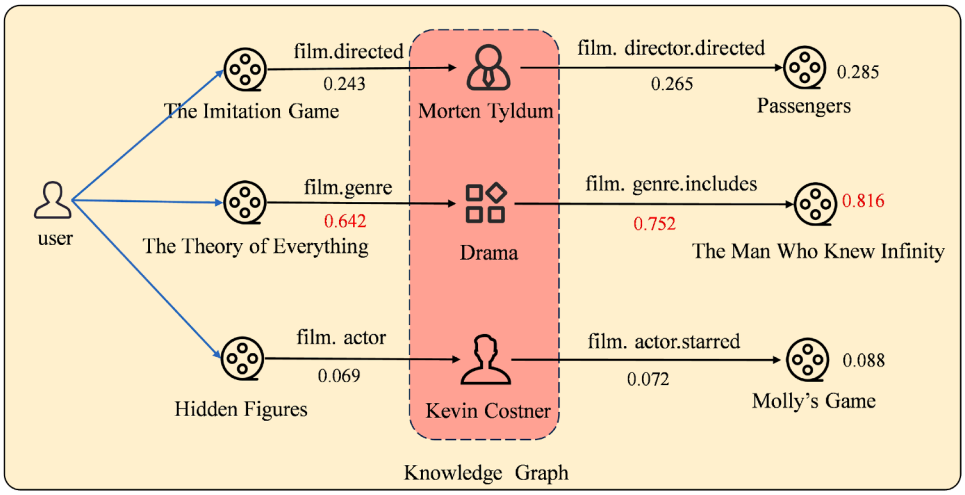



**Fig. 11.** Visualization of the preference evolution reasoning process in the knowledge graph.

tion. To illustrate how the Preference Evolution Attention mechanism shapes the embedding space, we visualize the t-SNE distributions of user (blue) and item (pink) embeddings before and after training in Fig. 10a and b.

As shown in Fig. 10a, before training the embeddings exhibit a dispersed and unstructured distribution. User and item embeddings are loosely scattered without clear clustering patterns, indicating weak semantic alignment. At this stage, the model has not yet learned meaningful semantic correspondence between users and items, and the interaction signals remain insufficiently represented.

In contrast, Fig. 10b shows that after training the embeddings evolve into distinct and compact clusters. Within these clusters, user and item embeddings become closely intermixed, demonstrating that the model effectively brings users closer to the items that match their preferences. This more compact embedding structure reflects stronger user-item correlations and improved semantic consistency, enabling the model to generate more accurate and personalized recommendations.

### 5.7.2. *Preference reasoning visualization*

To better illustrate the reasoning process of DPRec, we visualize a real recommendation case based on a local knowledge subgraph, as shown in Fig. 11. A user has interacted with several movies, including *The Imitation Game*, *The Theory of Everything*, and *Hidden Figures*. DPRec encodes these interactions and propagates preference signals along multi-hop semantic paths in the KG. Through the Preference Evolution Attention mechanism, the model evaluates the importance of different entities and relations, assigning adaptive weights to directors, genres, and actors. For this user, the relation `film.genre` receives the highest attention weight of 0.642, indicating a strong preference for movie genres, particularly *Drama*. Along the diffusion path `film.genre` → `Drama` → `film.genre.includes`, DPRec identifies that *The Man Who Knew Infinity* shares similar genre semantics with *The Theory of Everything*, which the user has previously watched. Consequently, the model recommends this film and provides the corresponding reasoning path as transparent evidence for the recommendation. This case demonstrates that DPRec not only captures explicit user-item correlations but also uncovers latent semantic relationships, resulting in explainable, semantically consistent, and personalized recommendations.

## 6. Conclusion and future work

In this paper, we proposed DPRec, a dual-path preference evolution framework with semantic alignment for KG recommendation. The model was designed to address the limitations of existing KG-based recommendation methods, which often suffer from preference dilution, noise propagation, and insufficient semantic alignment between user behaviors and relational structures. DPRec leverages a dual-path preference diffusion mechanism to achieve fine-grained modeling of user's direct and indirect preferences, while the preference evolution attention aggregation mechanism adaptively captures dynamic preference changes during the propagation process. Furthermore, the multi-level contrastive learning framework promotes effective alignment between preference representations and path-level semantics. More importantly, the primary innovation of this work lies in the proposed collaborative multi-module architecture. The dual-path preference diffusion, preference evolution attention aggregation, and multi-level contrastive learning modules are integrated into a unified framework, where they collaboratively enhance the joint representation capability of structural features and semantic information from three perspectives, namely preference modeling, dynamic evolution, and semantic alignment.

Comprehensive experiments on multiple benchmark datasets demonstrate that DPRec consistently outperforms representative GNN- and KG-based baseline methods in terms of recommendation accuracy, interpretability, and generalization capability, achieving statistically significant improvements on most evaluation metrics. These results verify that the unified modeling strategy adopted by DPRec effectively improves fine-grained preference modeling and enhances the reliability of recommendation results in complex heterogeneous environments.

Although DPRec achieves promising performance for KG recommendation, several limitations remain and provide directions for future research. First, the current framework mainly relies on user–item interaction information and knowledge graph structures, while richer multimodal semantic information, such as textual, visual, and social information, is not fully exploited. Future work could incorporate multimodal data to further enhance user preference understanding and improve model interpretability. Second, DPRec is mainly designed for single-domain recommendation scenarios, and its capability of cross-domain preference knowledge transfer remains limited. Future studies could investigate more effective cross-domain preference transfer strategies to improve the adaptability of the model in heterogeneous recommendation environments. Finally, the dual-path diffusion and contrastive learning mechanisms introduce additional computational overhead, which may affect scalability in large-scale recommendation scenarios. Therefore, developing more efficient diffusion and semantic alignment strategies to improve the scalability of DPRec in real-world applications will be an important direction for future research.

## CRediT authorship contribution statement

**Weihua Xu:** Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization; **Linxi Yin:** Writing – original draft, Visualization, Software, Investigation, Formal analysis, Data curation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

Our implementation code can be accessed at the following address: https://github.com/pickle233/DPRec.

## Appendix A. List of mathematical notations

To facilitate the understanding of the proposed DPRec framework, we provide a summary of the main mathematical notations used throughout this paper. Table A.1 lists the symbols and their corresponding descriptions, including user/item representations, knowledge graph components, contrastive learning objectives, and optimization parameters.

**Table A.1**
Summary of main mathematical notations.

| Symbol | Description |
|---|---|
| $\mathcal{U}$ | The set of users |
| $\mathcal{I}$ | The set of items |
| $\mathbf{Y}$ | User-item interaction matrix |
| $\mathcal{G}$ | Knowledge graph |
| $\mathcal{E}$ | The set of entities in the knowledge graph |
| $\mathcal{R}$ | The set of relations in the knowledge graph |
| $\mathcal{T}$ | The set of triples in the knowledge graph |
| $\phi(\cdot)$ | Mapping function from items to KG entities |
| $\mathcal{E}_u^{\text{dir}}$ | Direct preference entity set of user $u$ |
| $\mathcal{E}_u^{\text{ind}}$ | Indirect preference entity set of user $u$ |
| $\mathcal{T}_x^{v,(l)}$ | Triple set sampled for entity $x$ at layer $l$ under view $v$ |
| $\mathbf{X}_u$ | User interaction feature matrix of user $u$ |
| $\mathbf{h}, \mathbf{r}, \mathbf{t}$ | Head-entity, relation, and tail-entity embeddings |
| $\mathbf{p}_u^{(0)}$ | Initial preference representation of user $u$ |
| $\mathbf{e}_x^{(l)}$ | Representation of user/item $x$ at layer $l$ |
| $\mathbf{e}_x^{\text{dir}}$ | Aggregated direct-view representation |
| $\mathbf{e}_x^{\text{ind}}$ | Aggregated indirect-view representation |
| $\mathbf{Path}_x^{(l)}$ | Path-level representation of entity $x$ at layer $l$ |
| $\mathcal{L}_{\text{pref}}$ | Preference-level contrastive loss |
| $\mathcal{L}_{\text{path}}$ | Path-level contrastive loss |
| $\mathcal{L}_{\text{rec}}$ | Recommendation loss based on BPR |
| $\tau$ | Temperature coefficient in contrastive learning |
| $\lambda_1, \lambda_2, \lambda_3$ | Regularization and contrastive-loss weighting coefficients |
| $L$ | Number of propagation layers |
| $K$ | Sampling size at each propagation layer |
| $N$ | Size of the candidate embedding pool |
| $B$ | Mini-batch size |
| $d$ | Embedding dimension |

## Appendix B. Statistical significance test and effect-size analysis

To evaluate the statistical reliability of DPRec, we conduct statistical significance tests and effect-size analyses based on five independent runs using the fixed random seeds {2022, 2023, 2024, 2025, 2026}. Specifically, a two-sided paired $t$-test is performed to assess the statistical significance of the performance differences between DPRec and the strongest baseline, while Cohen's $d_z$ is calculated to quantify the corresponding effect size. The detailed results are reported in Table B.1.

**Table B.1**
Statistical significance test and effect-size analysis of DPRec compared with the strongest baseline.

| Dataset | Metric | $p$-value | Cohen's $d_z$ |
|---|---|---|---|
| Music | AUC | $1.36 \times 10^{-3}$ | 3.53 |
| | F1 | $4.95 \times 10^{-4}$ | 4.69 |
| | Hit@1 | $1.09 \times 10^{-3}$ | 3.76 |
| | Hit@5 | $1.21 \times 10^{-4}$ | 6.07 |
| | Hit@10 | $6.63 \times 10^{-4}$ | 4.28 |
| | NDCG@1 | $1.09 \times 10^{-3}$ | 3.76 |
| | NDCG@5 | $3.35 \times 10^{-3}$ | 2.78 |
| | NDCG@10 | $2.14 \times 10^{-3}$ | 3.13 |
| Book | AUC | $3.96 \times 10^{-2}$ | 1.35 |
| | F1 | $7.87 \times 10^{-3}$ | 2.21 |
| | Hit@1 | $3.13 \times 10^{-3}$ | 2.85 |
| | Hit@5 | $1.23 \times 10^{-3}$ | 3.64 |
| | Hit@10 | $8.99 \times 10^{-3}$ | 2.12 |
| | NDCG@1 | $3.13 \times 10^{-3}$ | 2.85 |
| | NDCG@5 | $2.65 \times 10^{-2}$ | 1.53 |
| | NDCG@10 | $1.41 \times 10^{-3}$ | 3.50 |
| Movie | AUC | $2.82 \times 10^{-4}$ | 5.37 |
| | F1 | $9.60 \times 10^{-4}$ | 3.85 |
| | Hit@1 | $1.61 \times 10^{-2}$ | 1.79 |
| | Hit@5 | $1.04 \times 10^{-3}$ | 3.68 |
| | Hit@10 | $7.50 \times 10^{-1}$ | 0.15 |
| | NDCG@1 | $1.61 \times 10^{-2}$ | 1.79 |
| | NDCG@5 | $2.01 \times 10^{-4}$ | 5.65 |
| | NDCG@10 | $7.28 \times 10^{-3}$ | 2.25 |

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